Control Is Motion

Geometry, Feasible Dynamics, and Robust Outcomes

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Chapter 1

Motion, Targets, and Timing

A golf swing is a motion undertaken for an outcome. The club must arrive at contact with a useful position, velocity, orientation, and impact location; the subsequent collision and flight determine what the ball does. The body must produce that delivery through a feasible movement, tolerate relevant variation, and continue into follow-through. Studying only a static pose misses much of this problem. Discarding the outcome would miss its purpose.

"Control is motion" is the organizing perspective of this book. It directs attention to how a trajectory is generated, shaped, and made reliable. It is not a claim that trajectory tracking, optimal control, or orbital stability replaces classical control theory. Those established tools already address moving references, periodic behavior, disturbances, and constrained objectives. The task is to choose and combine them correctly for the question at hand.

The important connections are physical. Earlier applied work changes the state available later. Configuration and velocity change inertial coupling, force transmission, and external loads. Grip and ground constraints restrict compatible motion and affect reactions. Timing changes velocity and effort even along an unchanged configuration path. Feedback and constitutive response alter how disturbances develop. Impact samples the resulting state at an event, which need not occur at a fixed clock time. A rigorous account keeps these connections while distinguishing a model identity from a claim about a golfer.

1.1 Specify What the Motion Should Produce

Let the state be x, the effective input be u, and the declared dynamics be

$$\dot{x} = f(x, u, t), \qquad u \in \mathcal{U}(x, t). \tag{1.1}$$

The state must contain enough information for the model to predict its own continuation. For a rigid mechanical model it includes configuration and velocity. A muscle-driven or flexible model may also need activation, tissue, shaft, and other internal states. A list of joint angles alone is not generally a complete dynamic state. Contact modes and their transition rules are part of the model as well.

Several objectives are useful and can coexist. Regulation keeps a quantity near a desired constant value. Trajectory tracking compares the state or output with a time-indexed reference. Path following permits progress along a geometric path with a separately specified timing objective. Terminal or event objectives constrain the state delivered at a particular time or crossing. Periodic tasks may call for orbital stabilization. The names describe different specifications; they are not mutually exclusive schools of control.

For a simplified impact investigation, let the first intended approach crossing be described by

$$h(x(t_e)) = 0, \qquad \nabla h(x(t_e))^T f(x(t_e), u(t_e), t_e) \neq 0, \tag{1.2}$$

with an appropriate approach direction and contact mode. The nonzero derivative is a transversality condition: it excludes a grazing crossing where event time can be especially sensitive. Define an output $y_e = H(x(t_e^-))$ from the pre-impact state and a target set $\mathcal{Y}$ of acceptable deliveries. Its components might describe clubhead velocity, face orientation, path direction, and impact location. A ball-outcome model can map those quantities, together with ball and contact properties, into launch and flight predictions.

A terminal requirement $y_e \in \mathcal{Y}$ is compatible with a moving club. It does not demand convergence to an equilibrium with nonzero velocity, nor require that the club accelerate through the collision. The impact model must account for the rapid change in momentum. Follow-through and any subsequent constraints require their own continuation. The event model, output definition, and admissible set determine what constitutes success.

This distinction is practical: two motions can have similar joint paths but different delivery speeds, while different joint paths can deliver similar club states. Neither observation by itself identifies how individual muscles produced the motion.

1.2 A Moving Reference Must Be Feasible

A nominal time-indexed trajectory $x_d(t)$ must have a compatible admissible input $u_d(t)$:

$$\dot{x}_d(t) = f(x_d(t), u_d(t), t), \qquad u_d(t) \in \mathcal{U}(x_d(t), t). \tag{1.3}$$

Joint limits, closure conditions, contact forces, and internal-state dynamics must also hold. Drawing a smooth curve in state space does not establish these properties. If a tracking error is expected to remain zero, the closed-loop vector field must reproduce the nominal tangent there.

For a control-affine model $f = f_0 + Gu$, define $r = \dot{x}_d - f_0(x_d, t)$. At a given nominal state, an unconstrained input exists only if $r \in \operatorname{im} G$. When that condition holds,

$$u_d = G^\dagger r + (I - G^\dagger G)z \tag{1.4}$$

expresses the family of solutions in chosen coordinates. The pseudoinverse selects a minimum Euclidean-norm representative; that criterion depends on input scaling and is not a muscle-recruitment law. Input bounds can eliminate all members of the family. A nonlinear input map may require a different solution procedure and still need not have an inverse.

For example, $G = (1, 0)^T$ cannot produce $r = (0, 1)^T$. Its least-squares solution leaves a nonzero residual. Conversely, $G = (1, 1)$ with $r = 1$ admits both $(1, 0)^T$ and $(1/2, 1/2)^T$, among infinitely many choices. Feasibility, uniqueness, and the selection of a preferred input are separate questions.

One useful architecture is $u = u_d + \Delta u$, with corrections designed for the actual tracking-error dynamics. It does not imply a universal feedforward share of measured joint torque. A nominal passive motion can have $u_d = 0$, whereas a strongly forced motion may require substantial nominal input. The split also depends on the chosen reference and policy representation. Establishing a neural feedforward or feedback contribution requires an experimental and physiological model beyond this algebraic split.

1.3 Tracking and Regulation

In a local Euclidean chart, set $e = x - x_d(t)$. Then

$$\dot{e} = f(e + x_d(t), u, t) - \dot{x}_d(t). \tag{1.5}$$

Tracking becomes regulation of the origin in a generally time-varying error system. This change of variables explains why Lyapunov methods and linear-quadratic designs remain useful for moving references. The MIT treatment of finite-horizon LQR and trajectory stabilization provides an established development of these tools. Applying an equilibrium controller without accounting for the changing reference can fail; that is a modeling or design error, not a limitation of all regulation-based analysis.

For a scalar example, take $\dot{x} = x + u$ and $x_d(t) = \sin t$. The nominal input is $u_d = \cos t - \sin t$. Choosing

$$u = u_d - (k+1)e, \qquad k > 0, \tag{1.6}$$

gives $\dot{e} = -ke$ exactly. The state keeps moving while its tracking error converges. If actuator bounds are imposed, this result applies only where the specified input can actually be delivered. Delays, unmodeled dynamics, and measurement error also require their own treatment.

1.4 Configuration Paths, Timing, and State

Write a configuration path as $q_d(s)$, where s is a dimensionless progress coordinate. For a timing law $s(t)$, define $\sigma = \dot{s}$. The chain rule gives

$$\begin{aligned} q(t) &= q_d(s(t)), \\ \dot{q}(t) &= q_d'(s)\sigma, \\ \ddot{q}(t) &= q_d''(s)\sigma^2 + q_d'(s)\dot{\sigma}. \end{aligned} \tag{1.7}$$

Thus the mechanical state is $(q_d(s), q_d'(s)\sigma)$, not a configuration-only curve. Changing σ changes that state and usually the required loads. A prescribed full-state curve $(q_d(s), v_d(s))$ has an additional compatibility condition $q_d'(s)\sigma = v_d(s)$. It cannot generally be traversed at any chosen speed while remaining the same physical state trajectory.

For a mechanical model $M(q)\ddot{q} + h(q, \dot{q}) = B(q)u$, substitution yields

$$B(q_d)u = M(q_d)\big(q_d''\sigma^2 + q_d'\dot{\sigma}\big) + h(q_d, q_d'\sigma). \tag{1.8}$$

The right side must lie in the admissible input image. Closure reactions, external forces, and constitutive loads must be included consistently in a more complete model. Geometric path design and timing design are useful subproblems, but the dynamics couples them. Underactuation makes that compatibility especially important; it does not make arbitrary retiming free.

As a reproducible example, take $m\ddot{q} = u$ with mass $m = 1$ kg and applied force u. Let $q_d(s) = \ell s^2$ with $\ell = 1$ m. Use $s = t/T$ on $0 \le t \le T$. The same configuration path from 0 to 1 metre gives

$$q = \ell(t/T)^2, \qquad \dot{q} = 2\ell t/T^2, \qquad u = 2m\ell/T^2. \tag{1.9}$$

At the endpoint, $T = 1$ s gives speed 2 m/s, force 2 N, and kinetic energy 2 J. With $T = 2$ s, the values are 1 m/s, 0.5 N, and 0.5 J. Work over the one-metre displacement equals the kinetic-energy increase in each case. The phase-plane curves are $\dot{q} = 2\sqrt{\ell q}/T$, so they are different state-space curves. This is an illustrative timing calculation, not a measured golf swing.

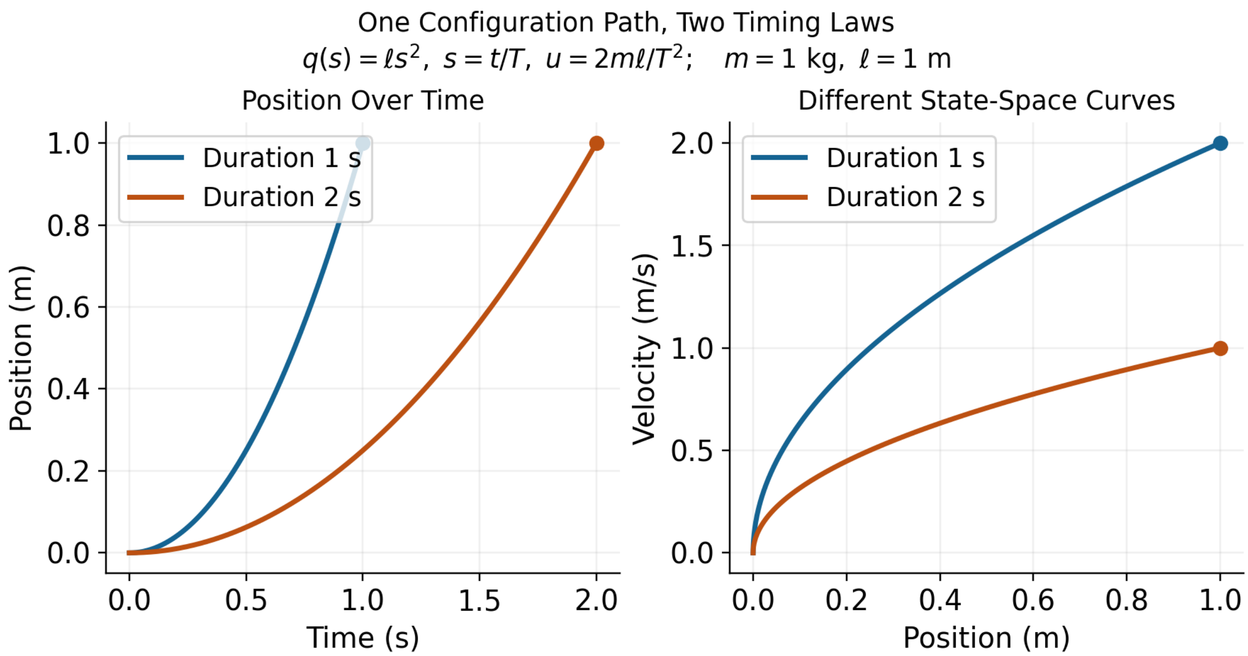


Figure 1.1: One Configuration Path, Two Timing Laws, and Different State Trajectories. Curves End at the Same Position With Different Velocities.

The example uses a prescribed clock-based phase. Estimating phase from the configuration itself needs care near the stationary start, where $q_d'(0) = 0$. In a swing, a coordinate that reverses at the top of the backswing cannot serve as a globally increasing phase through that reversal without a different construction. A useful phase must be defined on the intended interval, be observable from the chosen state or measurements, and have the required regularity and progress properties.

1.5 Choose a Meaningful Metric and Phase

A raw norm of position and velocity coordinates mixes different units. Even among positions, changing metres to centimetres changes an unweighted norm. One local choice is

$$\|e\|_W^2 = e^T W e, \qquad W = W^T \succ 0, \tag{1.10}$$

with weights and scales declared according to the desired interpretation. Under an invertible linear coordinate change $\widetilde{e} = Se$, the same quantity uses $\widetilde{W} = S^{-T} W S^{-1}$. Keeping W unchanged while changing units generally changes the specification. A physically motivated metric or scaled output error can be useful, but neither is automatically a clinical risk measure or an empirical measure of human effort.

On a configuration manifold, subtraction is a local coordinate operation; rotations, for example, need compatible charts or a chosen local logarithmic error. The mass matrix defines a kinetic-energy metric on configuration velocities, not by itself a unique distance on the entire augmented state including angles, velocities, activation, and other quantities.

For an embedded regular reference curve $\gamma(s)$ in a scaled state chart, one possible local phase is

$$s_*(x) \in \arg\min_s \frac{1}{2}(x - \gamma(s))^T W (x - \gamma(s)). \tag{1.11}$$

At an interior minimizer with constant W,

$$\gamma'(s_*)^T W (x - \gamma(s_*)) = 0. \tag{1.12}$$

A sufficiently small tubular neighborhood of a suitably regular embedded curve can make the projection unique and smooth. That conclusion is local. At the center of a circle every point on the circle is equally close; self-intersections, nearby branches, zero tangents, or a finite arc's endpoints can invalidate the assumed phase chart. A constrained endpoint minimizer need not satisfy the interior orthogonality equation.

Transverse coordinates need not use nearest-point orthogonality, but must form valid local coordinates together with phase. If the state dimension is d, there are locally $d - 1$ transverse coordinates around a regular one-dimensional curve. This coordinate count is different from the number of actuators and from the dimension of a task-equivalent set.

1.6 Stability, Attraction, and Finite Motion

Lyapunov stability means sufficiently small initial errors remain small over the stated time domain. It does not by itself mean convergence. Asymptotic stability adds attraction as time tends to infinity. Exponential stability adds a quantitative decay bound with specified constants. These distinctions apply to equilibria and have corresponding trajectory and orbital formulations; see the MIT Lyapunov analysis notes.

For a periodic orbit of an autonomous closed-loop system, orbital stability allows phase displacement while controlling distance to the orbit. A nearby solution can converge to the orbit without converging to the same clock phase as a nominal solution. This is useful for periodic motions when phase locking is not the objective. It is not a general license to change the physical speed along a prescribed position-velocity curve.

A golf swing studied only up to an impact event is a finite-duration motion. Its appropriate claims may concern bounded deviation, successful event arrival, and delivered output tolerances. An infinite-time attraction statement needs a defined continuation, such as a periodic task or a subsequent stabilization phase. A finite reference arc is not an invariant orbit once the motion leaves its endpoint. Bounds on a finite interval should not be renamed asymptotic stability without that missing structure.

For example, $\dot{e} = e$ gives $e(t) = e(0)e^t$. On a fixed interval $[0, T]$, choosing $|e(0)| < \varepsilon e^{-T}$ keeps $|e(t)| < \varepsilon$. This finite-time bound holds even though errors amplify and the origin is unstable on the infinite time axis. A useful swing robustness result should therefore report the permitted initial/measurement uncertainty, the amplification or decay, the time or event interval, and the terminal output bound.

Orbital analysis still uses Lyapunov functions. For an explicit example, consider $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1 + u$. The unit circle is a nominal orbit with $u_d = 0$. Put $r^2 = x_1^2 + x_2^2$ and choose

$$u = kx_2(1 - r^2), \qquad V = \tfrac{1}{4}(r^2 - 1)^2, \qquad \dot{V} = -kx_2^2(r^2 - 1)^2 \le 0, \quad k > 0. \tag{1.13}$$

On a sufficiently small compact annulus around the unit circle, the largest invariant subset of $\dot{V} = 0$ is the circle itself. Points with $x_2 = 0$ away from the circle do not remain there, since $\dot{x}_2 = -x_1 \neq 0$; the origin is excluded by the annulus. Lyapunov and invariance arguments then establish local orbital attraction and stability. They do not establish phase locking or global attraction from the origin. The MIT limit-cycle treatment and Manchester's transverse-dynamics analysis provide the broader framework.

1.7 Verifying a Motion Tube

A time-varying tube can be defined by

$$\mathcal{T}(t) = \{x : V(x,t) \le \rho(t)\}. \tag{1.14}$$

For example, V may measure weighted tracking error. Selecting $\rho(t)$ to be small near impact expresses a desired tolerance. It does not show that the closed-loop motion can stay inside the tube. Even $\dot{e} = 0$ leaves a shrinking tube when initialized on its boundary: constant error cannot fit inside an arbitrarily decreasing allowed radius.

Suppose the closed-loop dynamics is well posed with admissible inputs, V and ρ are continuously differentiable, and the tube has a regular boundary on the domain of interest. For disturbances d in a declared set $\mathcal{D}$, a robust inward-boundary condition is

$$\partial_t V + \nabla_x V^T f(x, k(x,t), d, t) \le \dot{\rho}(t) \quad \text{when } V(x,t) = \rho(t), \quad \forall d \in \mathcal{D}. \tag{1.15}$$

Together with the appropriate regularity and initial membership, this supports forward invariance over the specified interval. The derivative must include reference motion and changing metric weights. If V and ρ are indexed by phase, their derivatives involve the actual $\dot{s}$; it cannot silently be replaced by one. Nonsmooth profile corners or hybrid resets need their own one-sided or transition conditions.

A scalar example makes the effort and disturbance limit explicit. Let $\dot{e} = -ke + d$, $|d| \le \bar{d}$, and require $|e| \le r(t)$. On either boundary, a sufficient condition is

$$\dot{r} \ge -kr + \bar{d}. \tag{1.16}$$

For $k = 2$, $\bar{d} = 0.1$, and $r(0) = 1$ in declared scaled units, the tight comparison radius is $r(t) = 0.05 + 0.95e^{-2t}$. Constant worst-direction disturbances attain its two boundaries. The nonzero limiting radius follows from the disturbance bound and available correction rate. A drawing that shrinks to zero would not certify this disturbed system.

For an impact specification, also require that every relevant pre-impact boundary state allowed by the verified tube maps into $\mathcal{Y}$, and that the intended event is reached. A phase estimate bounded below by $\dot{s} \ge \alpha_{\min} > 0$ can supply a progress bound on an appropriate interval; distance to a path alone cannot. Actuator saturation, sensor delay, contact feasibility, and uncertainty in the model must be included in the claimed guarantee. A verified model tube is still a statement about that model and its declared uncertainty set, not automatic validation on golfers.

1.8 Input Rank and Reachability

For an n-coordinate mechanical model with $m < n$ effective inputs, rank limits instantaneous input acceleration directions. It does not imply that all feasible states lie on a surface of dimension $2m$. In a state model, drift and input directions evolve as the system moves; their effect over time can provide additional reachable directions. The underactuation chapter develops the associated feasibility and reachability questions.

A concrete counterexample uses two unit masses, one grounded spring, and one coupling spring, all with unit coefficients. In scaled coordinates,

$$\ddot{q} + Kq = bu, \qquad K = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{1.17}$$

There is one applied force. With $x = (q_1, q_2, \dot{q}_1, \dot{q}_2)^T$,

$$\dot{x} = Ax + gu, \qquad A = \begin{pmatrix} 0 & I \\ -K & 0 \end{pmatrix}, \qquad g = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}. \tag{1.18}$$

The controllability matrix $[g, Ag, A^2g, A^3g]$ has determinant -1 and rank four. With unconstrained signed inputs, this linear system is controllable in its full four-dimensional state space. Input limits restrict the size of reachable sets and time needed; they do not restore the asserted universal $2m$ dimension bound.

A single trajectory is itself a one-dimensional curve, but the union of reachable trajectories and the span of a curve's successive derivatives are different objects. For the same system with $u = 0$ and $x(0) = (1, 0, 0, 0)^T$, the matrix $[Ax, A^2x, A^3x, A^4x]$ at the initial state also has determinant -1. Thus even an unforced example has four independent time derivatives. At that regular point, smooth arc-length reparameterization preserves their rank. Actuator count therefore does not cap the number of independent Frenet-frame directions at $2m$ either.

The useful golf interpretation is more specific. Coupling can transmit and redistribute the effects of earlier work and present loads. Whether an effective wrist coordinate is unactuated, weakly actuated, or actively constrained is a modeling and measurement question. A rapid release does not prove zero wrist moment, and passive planar dynamics does not guarantee three-dimensional face squaring. Force, joint moment, mechanical power, and muscle activation must be kept distinct when testing such hypotheses.

1.9 Reliability and Performance

A path error, a delivery error, and a ball-dispersion measure are different costs. Choose the one that represents the study, then state how the others enter the model. Both regulation and trajectory problems can use cost functionals over an entire history; neither is restricted to a single instantaneous error value.

For a specified stochastic model and admissible policy π, one possible criterion is

$$\begin{aligned} J_\pi &= \phi(y_e) + \int_0^{t_e} \ell(x(t), u(t))\, dt, \\ \min_\pi \quad & \mathbb{E}[J_\pi] + \lambda \operatorname{Var}(J_\pi), \qquad \lambda \geq 0. \end{aligned} \tag{1.19}$$

The cost and its units, disturbance distribution, event definition, and constraint handling must be declared. The variance weight has corresponding units; it expresses a design preference rather than a universal biological constant. A low variance can accompany a consistently poor mean outcome, so repeatability alone is not sufficient. Expected cost, tail risk, target-set probability, and worst-case guarantees answer different questions. If the intended event is not reached by an allowed horizon, define a failure cost or constraint; silently discarding those trials changes the question.

A candidate policy may use feedback, a learned reference, or both. Comparing policies requires recomputing the motion, loads, event time, and output in each branch. Reusing a measured state or reaction history after the intervention changes the trajectory can invalidate the comparison. Experimental perturbations and held-out predictions can test a model; fitting one observed motion does not establish that the nervous system optimized the proposed criterion.

A practical golf investigation can proceed as follows:

1. Define the delivered output, its tolerance, and the ball/contact model used to connect that output to performance.

2. Declare the mechanical boundary, contact modes, state variables, input meaning, parameter uncertainty, and available measurements.

3. Construct compatible nominal states and inputs. Check closure, timing, actuator limits, work, and contact feasibility before optimizing a path.

4. Model disturbances and measurement/actuation delays. Compare finite-time or event-based output sensitivity and verify any proposed tube condition.

5. Test predictions on independent swings or perturbations, including uncertainty in event timing and the inferred loads.

6. Separate demonstrated mechanical identities, validated predictions, physiological interpretations, and proposed coaching or design hypotheses.

This is the role of the subsequent chapters: curve geometry describes motion; configuration manifolds describe compatible coordinates; transverse analysis and underactuation constrain candidate designs; optimization and robustness analysis evaluate them; learning and experiments test how well they transfer. The later golf case study is a modeling proposal unless supported by a complete specified model, reproducible results, and validation data. The framework's value comes from making the links testable.

1.10 Chapter Exercises

1. Specify an impact-event output set that distinguishes clubhead speed, face orientation, and impact location. Explain why a terminal target does not require stopping there.

2. Derive the error dynamics for the scalar tracking example, then impose an input bound and identify where the proposed law becomes infeasible.

3. Reproduce the one-mass timing example. Compare endpoint velocity, input, work, and the phase-plane curves for $T = 1$ and $T = 2$ seconds.

4. For a prescribed pair $(q_d(s), v_d(s))$, derive the compatibility condition on $\dot{s}$ and give an example where arbitrary retiming fails.

5. Rescale a position coordinate from metres to centimetres. Transport the error metric and verify that the weighted distance remains unchanged.

6. Show why nearest-point phase is not unique at a circle's center and why a finite arc's endpoint needs a separate projection condition.

7. Derive the scalar robust-radius condition and integrate its two worst-direction disturbance cases. Explain the nonzero limiting radius.

8. Verify both rank-four matrices in the spring example. Distinguish actuator rank, reachable-set dimension, and a trajectory's derivative span.

9. Verify the orbital Lyapunov derivative and the invariant-set argument. Explain why this does not prove phase locking or global convergence.

10. Formulate two different reliability objectives for a golf study. State what data would distinguish their predictions and what remains unidentifiable.

Chapter 2

Curves in State Space

2.1 What a Curve Describes

A golf swing has several useful geometric descriptions. The clubhead traces a curve in three-dimensional physical space. The joint configurations trace a curve in configuration space. Positions, velocities, and any additional internal states trace a curve in state space. These descriptions answer different questions. A bend in a plot of joint angle against angular velocity is not a bend of the clubhead in metres, and its curvature is not a force.

The purpose of this chapter is to make geometry useful without asking it to supply missing dynamics. We develop arc length, curvature, moving frames, local transverse coordinates, and timing constraints. We then connect these tools to physical acceleration, admissible control, and the impact task. Geometry specifies a motion's shape within a stated space and metric. Dynamics determines which timed motions and corrections are possible. Measurements determine whether a proposed model describes an actual swing.

Throughout the Euclidean calculations, the ambient space is $\mathbb{R}^d$ with a specified inner product. A prime denotes differentiation with respect to the displayed curve parameter; a dot denotes physical time. A regular curve has a nonzero first derivative on the interval under discussion. Configurations are denoted by q, full states by x, and a physical point such as the clubhead by p. We use s for arc length and λ for a parameter that need not measure distance.

2.1.1 Geometry Depends on the Space and Metric

Euclidean curvature is invariant under regular reparameterization and rigid changes of Cartesian frame. It is an extrinsic property of the embedded curve: it measures how the curve bends in its ambient space. Calling this "intrinsic" merely because it does not depend on timing confuses two uses of geometry. A straight segment and a circular arc of the same length have the same one-dimensional arc-length metric, although their ambient curvatures differ.

Changing units without transforming the metric changes the calculation. For a state $x = (q, \dot{q})$, a raw sum of squared positions and velocities has no universal physical meaning. One possible declared convention is $z = Sx$, where S contains inverse reference scales, followed by Euclidean geometry in the dimensionless coordinates z. Different scales generally give different curvature signatures. This is a modeling choice to report and test, not a coordinate-free marker of skill.

For a positive-definite metric W, the squared displacement is $dx^T W dx$. Under $z = Sx$, the same metric has matrix $\widetilde{W} = S^{-T} W S^{-1}$. A constant metric can be handled through a linear change to orthonormal coordinates. With a position-dependent metric, ordinary second derivatives are not sufficient: curvature uses the appropriate covariant derivative. The kinetic-energy matrix $M(q)$ defines a metric on configuration velocities; it is not automatically a metric on an augmented state containing velocities, activation, and contact variables.

2.2 Regular Parameters and Arc Length

Let $\gamma : I \to \mathbb{R}^d$ be a C^2 curve, and let $h(\lambda) = \|\gamma'(\lambda)\| > 0$. Its arc-length function and unit tangent are

$$s(\lambda) = \int_{\lambda_0}^{\lambda} h(\mu)\, d\mu, \qquad T(\lambda) = \frac{\gamma'(\lambda)}{h(\lambda)}. \tag{2.1}$$

Because $ds/d\lambda = h > 0$, the inverse function theorem gives a local regular arc-length parameter. In that parameter $\|\gamma'(s)\| = 1$ and $T = \gamma'(s)$. Replacing λ by a smooth increasing diffeomorphism changes the rate of traversal while preserving the oriented curve. A merely increasing function can have zero derivative and need not preserve regularity at every point.

For example, $(\cos t, \sin t)$ and $(\cos t^2, \sin t^2)$ trace the same circle on suitable intervals. The second time parameterization has zero velocity at $t = 0$. Arc-length formulas that divide by speed apply for $t > 0$; the circle's geometry can separately be extended through the starting point. A temporal stop is not by itself a geometric cusp or an infinite curvature.

A reversal is even more informative. In normalized coordinates, the physical path $p(t) = (t^2, 0)$ has zero curvature on either side of $t = 0$, while velocity vanishes and reverses there. The unit tangent defined by time is undefined at the reversal. The full mechanical state $x(t) = (t^2, 2t)$ has derivative $(2t, 2)$ and is regular at the same instant. Thus the physical point can stop while the state continues moving. There is no universal conclusion that the top of a backswing is the point of maximum curvature in any of these spaces.

If a regular configuration path is parameterized by λ and traversed with $\dot{\lambda} > 0$, its chosen-metric speed is $\dot{s} = \|q'(\lambda)\|\dot{\lambda}$. In particular, $\dot{\lambda}$ is not Cartesian speed unless the path is parameterized by physical arc length. This distinction matters when converting a speed limit into a timing law.

2.3 Curvature and Physical Acceleration

For a unit-speed C^2 Euclidean curve,

$$k(s) = \frac{dT}{ds}, \qquad \kappa(s) = \|k(s)\|. \tag{2.2}$$

Differentiating $T^T T = 1$ gives $T^T k = 0$. Where $\kappa > 0$, the principal normal is $N = k/\kappa$. At a point with zero curvature, k remains defined but this particular normal does not. Curvature vanishing on an entire connected regular interval makes T constant and the curve a straight segment. Vanishing at one point does not: (t, t^3) has zero curvature at $t = 0$ and bends arbitrarily nearby.

For any regular parameter, differentiating the normalized tangent gives

$$\kappa = \frac{\|(I - TT^T)\gamma''\|}{\|\gamma'\|^2} = \frac{\sqrt{\|\gamma'\|^2\|\gamma''\|^2 - (\gamma'^T\gamma'')^2}}{\|\gamma'\|^3}. \tag{2.3}$$

In three dimensions the numerator of the final expression is $\|\gamma' \times \gamma''\|$. In two dimensions it is the absolute determinant of the two derivative vectors. There is no need to invent an ordinary cross product in arbitrary dimension. The projected-acceleration form also avoids subtracting two large nearly equal squared quantities, although numerical error can still dominate near zero speed.

2.3.1 A Checkable Ellipse

For $\gamma(\lambda) = (a\cos\lambda, b\sin\lambda)$ with $a > b > 0$,

$$\kappa(\lambda) = \frac{ab}{(a^2\sin^2\lambda + b^2\cos^2\lambda)^{3/2}}. \tag{2.4}$$

The major-axis endpoints $(\pm a, 0)$, at $\lambda = 0, \pi$, have the largest curvature a/b^2. The minor-axis endpoints $(0, \pm b)$ have the smallest curvature b/a^2. With $a = 2$ m and $b = 1$ m these are 2 and 0.25 inverse metres. Uniformly changing the rate of traversal preserves these values. Stretching a circle into this ellipse does not preserve curvature.

2.3.2 Where the Speed-Squared Law Applies

Let $p(s)$ now be a physical position in an inertial Cartesian frame, with s measured in metres, and let $v = \dot{s} > 0$. The chain rule gives

$$\dot{p} = vT, \qquad \ddot{p} = \dot{v}T + v^2 k = \dot{v}T + v^2\kappa N \quad (\kappa > 0). \tag{2.5}$$

Here $v^2\kappa$ is a physical normal acceleration in m/s^2. The identity is kinematic: it describes the acceleration of the prescribed motion before assigning which forces produce it.

For a point mass, Newton's law says

$$m\ddot{p} = F_{\text{act}} + F_{\text{pass}} + F_{\text{external}} + F_{\text{reaction}}. \tag{2.6}$$

Only in a model where the actuator supplies the entire resultant force does $F_{\text{act}} = m(\dot{v}T + v^2 k)$. Gravity, springs, contact, and constraint reactions can contribute. A 0.2 kg mass moving at 20 m/s on a horizontal circle of radius 1 m requires 80 N inward. In an ideal frictionless circular guide with no tangential resistance, the guide supplies that force and does zero work because its force is perpendicular to velocity. No continuous actuator work is needed to maintain the speed in this model. That calculation does not show that a golfer supplies zero joint moments. It shows why resultant acceleration and active effort are different quantities.

A clubhead is a point on a moving, rotating, possibly flexible body rather than an isolated point mass with all force applied at that point. For a rigid model with $p = p(q)$,

$$\dot{p} = J_p(q)\dot{q}, \qquad \ddot{p} = J_p(q)\ddot{q} + \dot{J}_p(q, \dot{q})\dot{q}. \tag{2.7}$$

The Jacobian-rate term supplies normal acceleration even at constant joint speed. Segment force and moment balances, distributed mass, hand wrenches, and any modeled shaft deflection are needed to interpret measured clubhead acceleration. Multiplying it by total club mass does not generally recover the hand force.

For a first-order state equation $\dot{x} = f(x, u, t)$, the analogous curve identity concerns $\ddot{x}$, which may include mechanical jerk and internal state derivatives. When u is differentiable,

$$\ddot{x} = f_x f + f_u \dot{u} + f_t. \tag{2.8}$$

This is not a generic decomposition of u into tangential and normal forces. The reference condition remains $f(\gamma(s(t)), u_*(t), t) = \gamma'(s(t))\dot{s}(t)$, together with input admissibility. In an autonomous model the final argument is absent.

2.4 Moving Frames and Their Limits

2.4.1 An Oriented Three-Dimensional Frame

For a C^3 regular curve in oriented Euclidean three-space with $\kappa > 0$, define $T = dp/ds$, $N = T'/\kappa$, and $B = T \times N$. Then

$$\frac{dT}{ds} = \kappa N, \qquad \frac{dN}{ds} = -\kappa T + \tau B, \qquad \frac{dB}{ds} = -\tau N. \tag{2.9}$$

The signed torsion is

$$\tau = \frac{\det[p', p'', p''']}{\|p' \times p''\|^2}, \tag{2.10}$$

where the primes in this formula may use any regular parameter. Torsion requires nonzero curvature. A planar regular curve has zero torsion where the formula is defined; at an inflection the Frenet normal and torsion may be undefined even when the original curve is smooth.

For the helix $p(\lambda) = (a \cos \lambda, a \sin \lambda, b\lambda)$ with $a > 0$,

$$\kappa = \frac{a}{a^2 + b^2}, \qquad \tau = \frac{b}{a^2 + b^2}. \tag{2.11}$$

Changing the sign of b reflects the helix and changes the sign of torsion. Proper rotations and translations preserve both. Blindly applying Gram–Schmidt to all three derivatives chooses the final vector from the third derivative's residual; it can produce $-B$ when torsion is negative. That convention cannot simultaneously be identified with the oriented binormal $T \times N$ without tracking the sign.

In this physical three-dimensional setting, torsion measures rotation of the osculating plane per unit distance. Torsion of a normalized state-space curve has a different ambient space and interpretation. It is not a direct measure of anatomical rotation or of how a golfer should change the swing plane.

2.4.2 Higher Dimensions and Regular Normal Frames

With sufficient smoothness and independent successive derivatives, Gram–Schmidt constructs an orthonormal basis for the successive osculating spaces of a curve in $\mathbb{R}^d$. A full oriented Frenet frame can be chosen by orthogonalizing the first $d-1$ independent derivatives and completing the last vector with a fixed ambient orientation. Its equations have the form

$$e_i' = -\kappa_{i-1}e_{i-1} + \kappa_i e_{i+1}, \qquad \kappa_0 = \kappa_d = 0. \tag{2.12}$$

The first $d-2$ curvatures are positive under the required independence conditions; the final oriented curvature can have either sign or vanish. Alternatively, a frame made from all d independent derivatives adopts a different sign convention. State the convention before comparing signatures. If only a partial frame is constructed, its last derivative need not remain inside that partial span. A closed lower-dimensional Frenet system needs an appropriate containing affine subspace and its own regularity assumptions.

There is a simpler option for local control: choose any smooth orthonormal normal frame $E(s)$, with $d-1$ columns, satisfying $T^TE = 0$ and $E^TE = I$. Define

$$c = E^TT', \qquad \Omega = E^TE', \qquad T' = Ec, \qquad E' = -Tc^T + E\Omega. \tag{2.13}$$

Differentiating $E^TE = I$ shows that Ω is skew-symmetric. A parallel normal frame, often called a Bishop frame, chooses $\Omega = 0$ locally along an interval. It continues through zero-curvature points of a regular curve where the principal Frenet normal fails. Such a frame still requires an initial normal orientation; a periodic curve can have a nontrivial change of normal orientation after one circuit.

The number of normal coordinates is $d-1$, not the actuator count. The one-input spring example in the previous chapter has four independent unforced trajectory derivatives at its specified initial state. A single trajectory is one-dimensional, its osculating span can be larger, and the union of reachable trajectories is another object again.

2.5 Shape Reconstruction and Its Limits

For continuously differentiable $\kappa(s) > 0$ and continuous signed $\tau(s)$ on an interval, the three-dimensional Frenet equations, together with $p' = T$, reconstruct a unit-speed curve once its initial point and oriented frame are specified. Existence and uniqueness follow from the frame's linear differential equation and integration of T. Without the initial point and frame, the curve is determined up to a proper rigid motion. Analogous statements in higher dimensions require a complete consistent curvature list and frame convention; an arbitrary truncated signature is insufficient.

Congruence is not equality relative to the golf ball. Translating or rotating a clubhead path can make it miss the ball while preserving its curvature and torsion. A rigid rotation of normalized position-velocity coordinates need not even correspond to a physically allowed state transformation. Gravity, contact, anatomy, and the target frame further restrict useful symmetries. Neither shape reconstruction nor a similar curvature signature establishes equal timing, forces, face orientation, or performance.

Where $\kappa > 0$, the osculating circle has center $p + N/\kappa$ and radius $1/\kappa$. It matches local position, tangent, and curvature; the tangent line and osculating plane supply lower-order geometric descriptions. An osculating sphere in three dimensions needs additional conditions. Writing $\rho = 1/\kappa$, when $\tau \neq 0$ its center is $p + \rho N + (\rho'/\tau)B$. This follows by requiring successive derivatives of squared distance to a fixed center to vanish through third order. At zero torsion the construction may fail or be nonunique, depending on ρ'. It is not an unconditional approximation for every high-dimensional state curve.

2.6 Feasible Timing and Available Forces

For a chosen configuration path $q = q(\lambda)$,

$$\dot{q} = q'\dot{\lambda}, \qquad \ddot{q} = q''\dot{\lambda}^2 + q'\ddot{\lambda}. \tag{2.14}$$

Substitution into mechanical dynamics gives the required generalized load:

$$M(q)(q'\ddot{\lambda} + q''\dot{\lambda}^2) + h(q, q'\dot{\lambda}) = B(q)u + Q_{\text{pass}} + Q_{\text{external}} + J_c^T \eta. \tag{2.15}$$

Here h contains the declared velocity and potential terms, and passive terms must not be counted twice. The path must satisfy configuration constraints; its derivatives must satisfy their velocity and acceleration conditions. Contact reactions η, friction, actuator bounds, activation dynamics, and any force-rate limits constrain feasible timing. For an underactuated system, not every desired generalized load lies in the input image even if every motor is individually below its limit.

For a fully actuated model, each torque bound can become an inequality in λ, $\dot{\lambda}$, and $\ddot{\lambda}$. This is the basis of path-constrained time scaling. It is more informative than assigning a single curvature cost because it retains gravity, inertia, coupling, and the available input directions. The configuration path and timing can be optimized in stages, but their feasibility constraints remain coupled. Retiming also changes the corresponding full state curve, as shown in the previous chapter.

A simple point-mass model gives a useful limited speed bound. Suppose the actuator supplies the entire force, with $\|F\| \leq F_{\max}$, and there are no other forces. On a physical arc-length path,

$$\dot{v}^2 + (\kappa v^2)^2 \leq (F_{\max}/m)^2. \tag{2.16}$$

Thus $v \leq \sqrt{F_{\max}/(m\kappa)}$ is necessary when $\kappa > 0$, but leaves no force for tangential acceleration at equality. For $m = 2$ kg, $F_{\max} = 10$ N, $\kappa = 0.5\ \text{m}^{-1}$, and $v = 3$ m/s, the normal acceleration is 4.5 m/s^2 and the remaining tangential bound is $\sqrt{25 - 4.5^2} = \sqrt{4.75}$ m/s^2. If a guide supplies normal reaction, the actuator's feasible set is different.

At zero curvature this particular normal-acceleration bound supplies no speed ceiling. Tangential acceleration, initial and terminal conditions, velocity limits, power, and the next bend can still limit speed. A minimum-time exercise based only on normal acceleration can therefore lack an attained smooth solution or a physically meaningful optimum. Prescribe the full bounds and endpoint conditions before claiming an optimal law.

2.7 Local Phase and Transverse Dynamics

2.7.1 Coordinates in a Tubular Neighborhood

Let $\gamma(s)$ be a regular embedded unit-speed state curve in a chosen Euclidean metric, and choose the smooth normal frame $E(s)$ above. In a sufficiently small neighborhood of an interior segment, write

$$x = \gamma(s) + E(s)\xi. \tag{2.17}$$

Nearest-point phase satisfies $T^T(x - \gamma) = 0$ there. Local uniqueness requires more than this stationary equation: the curve must not intersect itself in the neighborhood, and the coordinate Jacobian must be invertible. The scalar factor $1 - c^T\xi$ in that Jacobian must be nonzero. A smaller neighborhood may be needed to exclude other closest points. For a circle, the center has many nearest points and the inward normal coordinates become singular there. Endpoints require one-sided treatment. A regular local normal frame does not turn all nearby states into a single global chart.

Let the smooth autonomous dynamics be $\dot{x} = f(x, u)$ and suppose a feasible nominal input satisfies $f(\gamma(s), u_*(s)) = v_*(s)T(s)$ with $v_* > 0$. Differentiate the coordinate map and project onto T and E:

$$\dot{s} = \frac{T^T f(x,u)}{1 - c^T\xi}, \qquad \dot{\xi} = E^T f(x,u) - \Omega\xi\dot{s}. \tag{2.18}$$

These equations retain both the phase denominator and the normal-frame transport. They are coordinate-rate equations; the word “force” should not be assigned to every transport term in a first-order state model.

With $u = u_*(s) + \delta u$, the transverse linearization at $\xi = 0$ is

$$\dot{\xi} = A_\perp(s)\xi + B_\perp(s)\delta u + O(\|(\xi, \delta u)\|^2), \tag{2.19}$$

where

$$A_\perp = E^T f_x(\gamma, u_*)E - v_*\Omega, \qquad B_\perp = E^T f_u(\gamma, u_*). \tag{2.20}$$

All partial derivatives here hold s fixed; nominal phase evolves by $\dot{s} = v_*(s)$. For explicitly time-dependent dynamics, include the time state or derive the corresponding time-dependent coordinate map. A feedback depending on an independently prescribed clock is also a different problem from the phase-indexed law assumed here.

There is no universal correction of the form speed squared times a curvature matrix subtracted from the original Jacobian. Curvature enters the phase map and chosen frame; transverse stability depends on the actual vector field, its derivatives, input directions, feedback, and regularity. Changing the normal basis changes the coordinate matrices consistently, without changing physical attraction of the orbit.

2.7.2 The Same Circle With Opposite Stability

In a neighborhood of $r = R > 0$, consider the planar systems

$$\dot{r} = a(r - R), \qquad \dot{\theta} = \omega > 0. \tag{2.21}$$

Every value of a has exactly the same nominal circle, curvature $1/R$, speed $R\omega$, and nominal time parameterization. Yet radial error evolves as $e(t) = e(0)e^{at}$: $a < 0$ attracts, $a > 0$ repels, and $a = 0$ is neutral. Curvature and nominal speed cannot distinguish these cases. The figure uses $R = 2$ and $\omega = 1.3$, with initial radii 1.8 and 2.2, over four model time units. Its radial growth rates are $a = -0.3$ and $a = 0.3$.

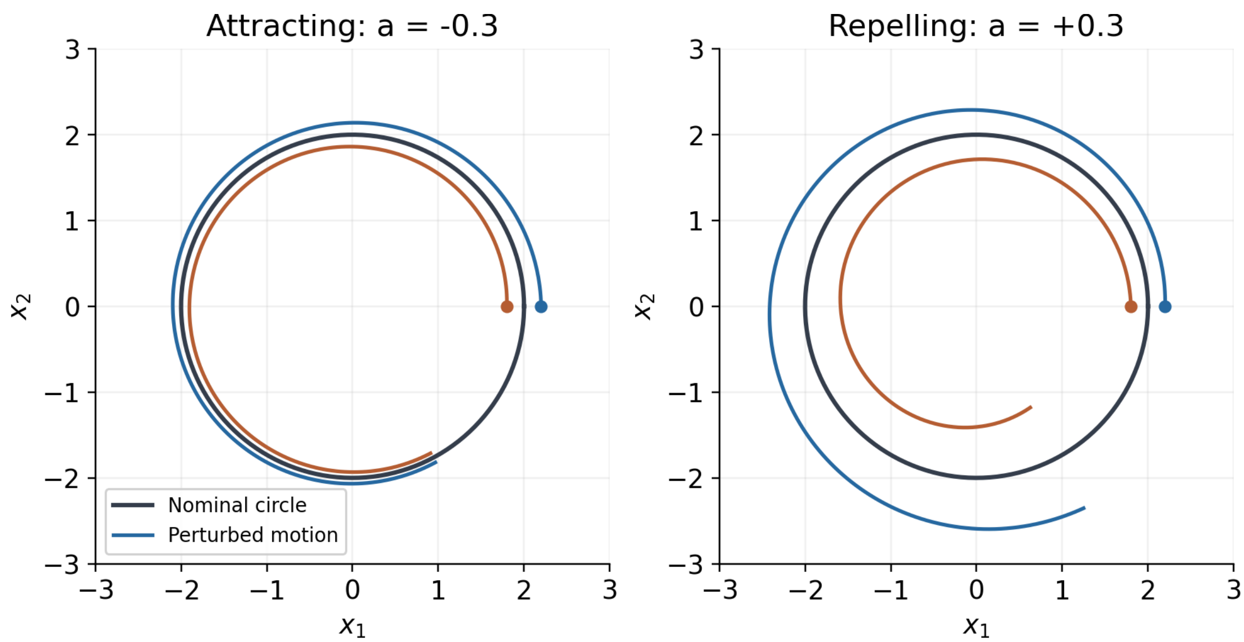


Figure 2.1: The Same Nominal Circle With Opposite Transverse Stability

Using the inward normal coordinate $\xi = R - r$ and arc length $s = R\theta$ gives $\dot{\xi} = a\xi$ and $\dot{s} = R\omega$. The exact phase formula agrees: $T^T f = \omega(R - \xi)$ and $1 - c^T\xi = 1 - \xi/R$. Their ratio is $R\omega$. This checks the denominator as well as the transverse linearization. A geometric tube around the circle becomes a verified invariant tube only when the actual transverse dynamics and disturbances satisfy its boundary condition.

2.8 Curvature and Reachability Are Different

For a control-affine system, instantaneous input directions, reachable sets, and the derivatives of one trajectory are distinct. Lie brackets describe noncommuting flows. With the convention $[g_i, g_j] = Dg_j\, g_i - Dg_i\, g_j$, the bracket can expose a displacement direction generated by suitable short sequences of controls. The Chow–Rashevskii result concerns bracket-generating driftless systems under appropriate smoothness and reversible control assumptions. The orbit theorem describes the geometry of orbits of vector fields and is not the same theorem. Allowing forward and backward field flows to define an orbit does not automatically describe forward-time reachability with drift, unilateral contact, or bounded controls.

A curved path need not use nonzero brackets at all. For the planar integrator $\dot{x} = u$, the two constant input fields commute. The control $u(t) = (-\sin t, \cos t)$ traces the unit circle from $(1, 0)$ with curvature one. Its acceleration comes from changing the input, while every bracket of the constant input fields is zero. Conversely, a system can have useful bracket directions without expressing them as a large curvature along one recorded trajectory. High curvature does not identify a commutator, a neural synergy, or an underactuated strategy.

2.9 Using Geometry in a Golf Investigation

A useful investigation keeps the chain of interpretation explicit: measurement, geometric description, feasible mechanics, uncertainty, and task outcome. Consider the following protocol.

1. Choose the object and frame. A clubhead point, shaft orientation, joint configuration, and complete state are different data products. Use a target-fixed physical frame for ball delivery. State how angles, rotations, velocities, and internal states are represented.

2. Report units and metric scales. For a state signature, repeat the calculation under plausible scaling choices. For three-dimensional clubhead curvature, use physical positions in consistent length units.

3. Fit derivatives with an explicit measurement model. Differentiation amplifies noise; torsion requires a third derivative and is especially fragile near small curvature or speed. Report sampling, smoothing, uncertainty, and sensitivity to filtering. Do not divide by near-zero denominators and then interpret numerical spikes as swing phases.

4. Align trials using a declared event or phase convention. Compare both geometric shape and timing; report where a reference curve's nearest projection becomes ambiguous. Treat impact and contact changes using one-sided smooth segments or a hybrid model.

5. Compute physical acceleration and mechanical load with the appropriate Jacobians, inertias, external loads, and constraints. Distinguish net joint moments from muscle recruitment and resultant force from work. Validate these quantities against independent measurements where possible.

6. Evaluate pre-impact speed, face orientation, path direction, strike location, and relevant ball outcomes with uncertainty. Test whether a geometric feature predicts outcomes beyond simpler measured variables on held-out trials. A correlation does not identify the control policy.

7. Test interventions in a declared forward model and then seek empirical validation. Altering timing can change speed-dependent loads and contact feasibility; altering a path can change moment arms and the impact frame. Recompute the complete trajectory rather than attaching a new conclusion to an unchanged force history.

The hypothesis that golfers exploit passive coupling remains worth testing. The geometric tools can identify repeatable patterns and organize model comparisons. They cannot by themselves establish that a particular pause, release, curvature peak, or low-dimensional signature is optimal, automatic, or a consequence of a particular neural strategy. Those arguments require mechanics and evidence at the next links in the chain.

2.10 Further Reading

The geometric conventions can be checked against Petersen's Classical Differential Geometry, especially its curve, Frenet, and parallel-frame treatments. Path-constrained timing and torque bounds are developed in Lynch and Park's Modern Robotics. For transverse dynamics and orbital stabilization, see Manchester's Transverse Dynamics and Regions of Stability for Nonlinear Hybrid Limit Cycles. These are mathematical and robotics references; they do not validate the golf-specific hypotheses in this chapter.

2.11 Exercises

1. For (t, t^3), derive the curvature and show why its zero value at the origin does not imply a straight neighborhood. Contrast this with the stopped physical path $(t^2, 0)$ and its regular position-velocity state.
2. Derive the ellipse formula and identify its major- and minor-axis endpoint curvatures. Verify invariance under $t \mapsto 2t$ and explain why stretching only one coordinate changes the Euclidean answer.
3. Compute curvature and signed torsion for (t, t^2, t^3) at $t = 0$ and $t = 1$. Explain what extra mechanical assumptions are needed to compare actuator effort along the two portions of the curve.
4. For the helix $(2\cos t, 2\sin t, t)$, calculate T, N, B, κ, τ. Reflect its third coordinate and track the sign of the oriented torsion.
5. Derive the exact normal-coordinate equations by projecting the derivative of $x = \gamma(s) + E(s)\xi$. Identify the coordinate singularity for a circle and verify the radial stability counterexample.
6. A 2 kg point mass is driven solely by a force of norm at most 10 N. Find the available tangential acceleration at speed 3 m/s on a radius-2 m circle. Repeat the interpretation when an ideal guide supplies all normal reaction.
7. Let $p(\lambda) = \ell(\lambda, \sin(\pi\lambda), 0)$ for $0 \le \lambda \le 2$, where $\ell = 1$ m. Derive the curvature and the conversion from $\dot{\lambda}$ to physical speed. Formulate a timing optimization with a total-acceleration bound, a finite speed bound, and zero initial and terminal speeds; do not assume that a bound on normal acceleration alone specifies a smooth minimum-time solution.
8. Starting with a declared double-pendulum model, generate a smooth swing-like segment and compare configuration, physical clubhead, and normalized state curvature. Repeat under different state scales and sampling noise. Report which features remain stable and which do not.
9. Construct two congruent physical paths with different positions relative to a fixed ball. Explain why their curvature signatures agree while their impact outcomes differ.
10. For a measured or simulated swing, propose one geometric hypothesis, its mechanical interpretation, a competing explanation, an independent validation measurement, and a criterion that could falsify it.

Chapter 3

Configuration Manifolds

The map is not the territory. The coordinate chart is not the manifold. Every time you write θ for an angle, you are lying—just a little.

3.1 The Lie That Coordinates Tell

In Chapter 2 we developed the geometry of curves in $\mathbb{R}^n$. This was a necessary starting point, but it was also a simplification. The state spaces of real mechanical systems are *not* $\mathbb{R}^n$.

Consider the simplest possible rotating joint: a hinge. Its configuration is an angle θ, which we habitually write as a real number. But an angle is not a real number. The configuration $\theta = 0$ and the configuration $\theta = 2\pi$ are *the same physical state*. No point in $\mathbb{R}$ has this property. The true configuration space of a hinge is the *circle* S^1—a one-dimensional manifold that is locally like $\mathbb{R}$ but globally has a completely different topology.

This is not pedantry. It is the source of real engineering failures.

Warning 3.1. The Gimbal Lock Catastrophe

If we represent orientation using three Euler angles $(\phi, \theta, \psi) \in \mathbb{R}^3$, we encounter **gimbal lock**: configurations where the representation degenerates and loses a degree of freedom. At $\theta = \pm\pi/2$, the axes for ϕ and ψ align, making the system appear to have only 2 DOF when it actually has 3.

This is not a hardware problem—it is a *coordinate singularity*, an artifact of forcing a non-Euclidean space (SO(3)) into Euclidean coordinates ($\mathbb{R}^3$). Apollo 13's inertial measurement unit experienced gimbal lock operationally. Every modern spacecraft avoids Euler angles for precisely this reason.

The lesson for control: **if you use the wrong coordinates, your controller will have singularities that the physical system does not.**

Key Idea 3.1. The Manifold Imperative

The configuration space of a mechanical system is a *smooth manifold* $\mathcal{Q}$—a space that is locally like $\mathbb{R}^n$ but may have nontrivial global topology. Control theory built on $\mathbb{R}^n$ must be replaced by control theory built on $\mathcal{Q}$ whenever:

(i) Joints allow full rotation (wrapping around S^1 or SO(3)).

(ii) The trajectory traverses a significant fraction of the configuration space.

(iii) Singularity-free representation is required for robustness.

For the golf swing—which involves large rotations in multiple joints simultaneously—all three conditions are met.

3.2 Smooth Manifolds: The Minimum You Need

We develop only as much manifold theory as the control engineer needs. The reader seeking a comprehensive treatment should consult Lee (2012) or Abraham & Marsden (1978).

Definition 3.1. Smooth Manifold

A **smooth manifold** $\mathcal{M}$ of dimension n is a set together with a collection of **coordinate charts** $\{(U_\alpha, \varphi_\alpha)\}$ such that:

(i) Each $U_\alpha \subseteq \mathcal{M}$ is an open set, and the U_α cover $\mathcal{M}$.

(ii) Each $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ is a homeomorphism onto an open subset of $\mathbb{R}^n$.

(iii) Where two charts overlap, the **transition maps** $\varphi_\beta \circ \varphi_\alpha^{-1}$ are smooth (C^∞) functions from $\mathbb{R}^n$ to $\mathbb{R}^n$.

The key intuition is: a manifold is a space where you can *locally* use coordinates, but no single coordinate system works *globally*. You need an atlas of overlapping charts, like a road atlas needs multiple pages to cover a country.

Example 3.1. The Circle S^1

The circle $S^1 = \{(\cos\theta, \sin\theta) : \theta \in \mathbb{R}\}$ is a 1-dimensional manifold. It cannot be covered by a single coordinate chart: any angle parameterization $\theta \in [0, 2\pi)$ has a discontinuity at $\theta = 0 = 2\pi$. We need at least two charts:

$$U_1 = S^1 \setminus \{(-1, 0)\}, \qquad \varphi_1(p) = \theta \in (-\pi, \pi), \tag{3.1}$$

$$U_2 = S^1 \setminus \{(1, 0)\}, \qquad \varphi_2(p) = \theta \in (0, 2\pi). \tag{3.2}$$

On the overlap $U_1 \cap U_2$, the transition map is either the identity or a shift by 2π—both smooth.

For a hinge joint, this means: no single angle variable θ can represent all configurations without a discontinuity. If the joint rotates past the discontinuity, any controller using that angle will experience a "jump"—a coordinate artifact, not a physical event.

3.2.1 Tangent Spaces and Tangent Vectors

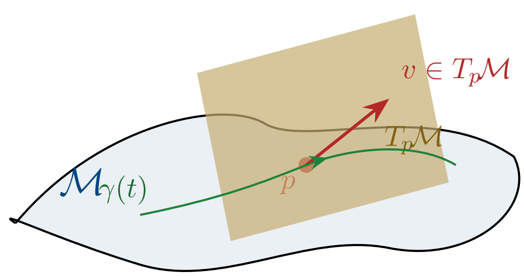
On a manifold, there is no global notion of "direction." Velocity is a *local* concept, defined at each point.

Definition 3.2. Tangent Space

The **tangent space** $T_p\mathcal{M}$ at a point $p \in \mathcal{M}$ is the vector space of all possible velocity vectors at p. If $\mathcal{M}$ has dimension n, then $T_p\mathcal{M} \cong \mathbb{R}^n$ as a vector space, but each point has its *own* tangent space.
Given a curve $\gamma : (-\varepsilon, \varepsilon) \to \mathcal{M}$ with $\gamma(0) = p$, the tangent vector $\dot{\gamma}(0)$ is an element of $T_p\mathcal{M}$.

Definition 3.3. Tangent Bundle

The **tangent bundle** $T\mathcal{M} = \bigsqcup_{p\in\mathcal{M}} T_p\mathcal{M}$ is the collection of all tangent spaces, one for each point. It is itself a smooth manifold of dimension $2n$. A point in $T\mathcal{M}$ is a pair (p, v) where $p \in \mathcal{M}$ and $v \in T_p\mathcal{M}$.
For a mechanical system with configuration manifold $\mathcal{Q}$, the **state space is the tangent bundle** $\mathcal{X} = T\mathcal{Q}$. A state is a pair $(\boldsymbol{q}, \dot{\boldsymbol{q}})$—a configuration and a velocity.



3.3 The Configuration Manifolds of Joints

Every mechanical joint constrains relative motion between two rigid bodies. The set of allowed relative configurations forms a manifold whose topology depends on the joint type.

Joint type	**DOF**	**Manifold**	**Topology**
Revolute (hinge)	1	S^1	Circle
Prismatic (slider)	1	$\mathbb{R}$	Line (or interval)
Universal	2	S^2	2-sphere
Ball-and-socket	3	$\mathrm{SO}(3)$	Rotation group
Planar	3	$\mathrm{SE}(3)\|_{2\mathrm{D}} = \mathbb{R}^2 \times S^1$	Plane + rotation
Free body	6	$\mathrm{SE}(3) = \mathbb{R}^3 \rtimes \mathrm{SO}(3)$	Rigid motions of 3-space

The configuration space of a multi-body system is the *product* of individual joint manifolds (modulo kinematic constraints):

Definition 3.4. Product Configuration Space

For a serial chain of k joints with individual configuration manifolds $\mathcal{M}_1, \ldots, \mathcal{M}_k$, the total configuration space is

$$\mathcal{Q} = \mathcal{M}_1 \times \mathcal{M}_2 \times \cdots \times \mathcal{M}_k. \tag{3.3}$$

For example, a robot arm with three revolute joints has $\mathcal{Q} = S^1 \times S^1 \times S^1 = \mathbb{T}^3$, the 3-dimensional torus.

Example 3.2. The Human Shoulder

The glenohumeral (shoulder) joint is approximately a ball-and-socket joint with configuration manifold $\mathrm{SO}(3)$. Combined with the 1-DOF hinge at the elbow (S^1) and the 1-DOF pronation/supination of the forearm (S^1), the arm has configuration space

$$\mathcal{Q}_{\text{arm}} = \mathrm{SO}(3) \times S^1 \times S^1. \tag{3.4}$$

This is a 5-dimensional manifold with the topology of $\mathrm{SO}(3) \times \mathbb{T}^2$. No set of 5 angle variables can parameterize this space without singularities.

3.4 The Rotation Group $\mathrm{SO}(3)$

The rotation group $\mathrm{SO}(3)$ appears wherever a joint allows full 3D rotation. It is the single most important non-Euclidean manifold in mechanical control, and its properties drive many of the challenges we will encounter.

Definition 3.5. The Special Orthogonal Group

SO(3) is the group of 3×3 real orthogonal matrices with determinant $+1$:

$$\mathrm{SO}(3) = \{\boldsymbol{R} \in \mathbb{R}^{3\times3} : \boldsymbol{R}^\top \boldsymbol{R} = \boldsymbol{I},\ \det(\boldsymbol{R}) = 1\}. \tag{3.5}$$

It is a 3-dimensional compact Lie group. As a manifold, it is diffeomorphic to the real projective space $\mathbb{R}P^3$.

3.4.1 Why Euler Angles Fail

The most common parameterization of SO(3) uses three Euler angles (ϕ, θ, ψ), defining the rotation matrix as a product of three elemental rotations. This parameterization has two fundamental problems:

P1. Singularities. Every 3-parameter representation of SO(3) has at least one singularity. For the ZYX convention, gimbal lock occurs at $\theta = \pm\pi/2$. This is a topological necessity, not a flaw in the choice of convention: SO(3) is not homeomorphic to any open subset of $\mathbb{R}^3$.

P2. Non-uniqueness. Euler angles are periodic: (ϕ, θ, ψ) and $(\phi + 2\pi, \theta, \psi)$ represent the same rotation. Near the singularity, drastically different angle triples can represent nearly identical orientations, creating numerical chaos in controllers.

Principle 3.1. Representation Principle for Rotations

For trajectory control on SO(3), use one of:

(a) **Rotation matrices $\boldsymbol{R} \in \mathbb{R}^{3\times3}$**: 9 parameters with 6 orthogonality constraints. Singularity-free but redundant.

(b) **Unit quaternions $\boldsymbol{q} \in S^3 \subset \mathbb{R}^4$**: 4 parameters with 1 unit-norm constraint. Singularity-free, minimal redundancy, excellent for interpolation and computation.

(c) **Exponential coordinates** (axis-angle or Lie algebra): 3 parameters with a singularity at $\|\boldsymbol{\omega}\| = \pi$, but natural for small rotations and linearization.

Never use Euler angles for control of large-rotation trajectories.

3.4.2 The Lie Algebra $\mathfrak{so}(3)$

The tangent space of SO(3) at the identity is the **Lie algebra** $\mathfrak{so}(3)$—the space of 3×3 skew-symmetric matrices:

$$\mathfrak{so}(3) = \{\boldsymbol{\Omega} \in \mathbb{R}^{3\times3} : \boldsymbol{\Omega}^\top = -\boldsymbol{\Omega}\} = \left\{ [\boldsymbol{\omega}]_\times := \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix} : \boldsymbol{\omega} \in \mathbb{R}^3 \right\}. \tag{3.6}$$

The "hat map" $(\cdot)^\wedge : \mathbb{R}^3 \to \mathfrak{so}(3)$ and its inverse "vee map" $(\cdot)^\vee : \mathfrak{so}(3) \to \mathbb{R}^3$ identify the Lie algebra with $\mathbb{R}^3$ (the space of angular velocity vectors).

Definition 3.6. Exponential Map

The **exponential map** $\exp : \mathfrak{so}(3) \to \mathrm{SO}(3)$ connects the Lie algebra to the Lie group:

$$\boldsymbol{R} = \exp([\boldsymbol{\omega}]_\times) = \boldsymbol{I} + \frac{\sin\theta}{\theta}[\boldsymbol{\omega}]_\times + \frac{1-\cos\theta}{\theta^2}[\boldsymbol{\omega}]_\times^2, \tag{3.7}$$

where $\theta = \|\boldsymbol{\omega}\|$. This is the **Rodrigues formula**. Geometrically, $\boldsymbol{R}$ is a rotation by angle θ about the axis $\boldsymbol{\omega}/\theta$.
The inverse map $\log : \mathrm{SO}(3) \to \mathfrak{so}(3)$ is well-defined for rotations with angle $\theta < \pi$ and gives the **rotation vector** representation.

The exponential map is the bridge between the linear world (the Lie algebra, where we can add and scale) and the nonlinear world (the Lie group, where the actual dynamics live). This bridge is the central tool for linearization on manifolds.

3.5 The Golf Swing Configuration Space

We now apply the manifold framework to our running example. The result will show exactly why Euclidean control theory is insufficient for the golf swing.

3.5.1 A Simplified Kinematic Model

Consider a 4-segment model of the golf swing:

(i) **Torso**: rotates relative to ground via the spine. Modeled as a 3-DOF ball joint: $\mathcal{M}_1 = \mathrm{SO}(3)$.

(ii) **Lead arm**: rotates relative to the torso via the shoulder. Modeled as a 3-DOF ball joint: $\mathcal{M}_2 = \mathrm{SO}(3)$.

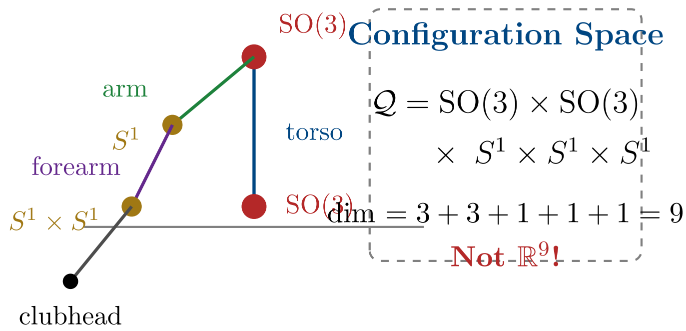
(iii) **Forearm/hands**: rotates relative to the upper arm via the elbow (1 DOF) and wrist flexion (1 DOF): $\mathcal{M}_3 = S^1 \times S^1$.

(iv) **Club**: rotates relative to the hands via wrist cock/uncock (1 DOF): $\mathcal{M}_4 = S^1$.

The total configuration space is:

$$\mathcal{Q}_{\text{golf}} = \mathrm{SO}(3) \times \mathrm{SO}(3) \times S^1 \times S^1 \times S^1. \tag{3.8}$$

This is a **9-dimensional manifold**. Its topology is $\mathrm{SO}(3) \times \mathrm{SO}(3) \times \mathbb{T}^3$—decidedly not $\mathbb{R}^9$.



3.5.2 Why This Matters for Control

The state space (tangent bundle) has dimension $2 \times 9 = 18$. A trajectory through this space is a curve on an 18-dimensional manifold. The consequences for control are immediate:

C1. No global linearization. You cannot write $\boldsymbol{x} = \boldsymbol{x}^* + \delta\boldsymbol{x}$ with $\delta\boldsymbol{x} \in \mathbb{R}^{18}$, because the "+" operation is not defined globally on the manifold. Linearization must be done in the *tangent space*, using the exponential map.

C2. Error is not a vector. The "error" between two configurations $\boldsymbol{q}_1, \boldsymbol{q}_2 \in \mathcal{Q}$ is not $\boldsymbol{q}_2 - \boldsymbol{q}_1$ (subtraction is undefined on manifolds). The correct notion of error is the *logarithmic map*: $\boldsymbol{e} = \log(\boldsymbol{q}_1^{-1}\boldsymbol{q}_2)$, which lives in the Lie algebra.

C3. Geodesics replace straight lines. The "shortest path" between two configurations is not a straight line in any coordinate system—it is a *geodesic* on the manifold, determined by the Riemannian metric (mass-inertia matrix).

C4. Parallel transport replaces vector addition. To compare tangent vectors (velocities, errors) at different points on the trajectory, we must *transport* them along the manifold using the connection. Simply subtracting velocity vectors at different times is mathematically meaningless.

Warning 3.2. Euler Angle Catastrophe in Golf Modeling

Many biomechanics studies parameterize the golf swing using Euler angles for each joint. For the shoulder joint (SO(3)), this means three angles $(\phi_s, \theta_s, \psi_s)$. But during the backswing, the shoulder passes through configurations near gimbal lock ($\theta_s \approx \pm\pi/2$).

In these regions:

- The Jacobian mapping angular velocities to Euler angle rates becomes singular.
- Small physical rotations produce large jumps in angle coordinates.
- Any controller computing torques from angle errors will produce *infinite* commanded torques at the singularity.

The fix is not to "choose a better Euler angle convention" (all conventions have singularities somewhere). The fix is to work on SO(3) directly, using rotation matrices or quaternions.

3.6 Riemannian Metrics and the Mass Matrix

A manifold by itself has no notion of distance, angle, or length. To measure these, we need a **Riemannian metric**—a smooth choice of inner product on each tangent space.

Definition 3.7. Riemannian Metric

A **Riemannian metric** on a manifold $\mathcal{M}$ is a smooth assignment of an inner product $\langle\cdot,\cdot\rangle_p$ on each tangent space $T_p\mathcal{M}$. In local coordinates $\boldsymbol{q} = (q_1, \ldots, q_n)$, the metric is represented by a positive-definite matrix $\boldsymbol{G}(\boldsymbol{q})$:

$$\langle \boldsymbol{v}, \boldsymbol{w} \rangle_{\boldsymbol{q}} = \boldsymbol{v}^\top \boldsymbol{G}(\boldsymbol{q})\, \boldsymbol{w}, \qquad \boldsymbol{v}, \boldsymbol{w} \in T_{\boldsymbol{q}}\mathcal{M}. \tag{3.9}$$

For mechanical systems, nature provides a canonical Riemannian metric: the **kinetic energy metric**.

Principle 3.2. The Mass Matrix as Riemannian Metric

For a mechanical system with generalized coordinates $\boldsymbol{q}$ and mass-inertia matrix $\boldsymbol{M}(\boldsymbol{q})$, the kinetic energy

$$T = \frac{1}{2}\dot{\boldsymbol{q}}^\top \boldsymbol{M}(\boldsymbol{q})\, \dot{\boldsymbol{q}} \tag{3.10}$$

defines a Riemannian metric $\boldsymbol{G} = \boldsymbol{M}(\boldsymbol{q})$ on $\mathcal{Q}$. This metric has deep physical significance:

(i) **Distance** between configurations is measured in “units of kinetic energy”—configurations connected by high-inertia motions are “far apart.”

(ii) **Geodesics** (shortest paths) under this metric are the trajectories of the *unforced* system—free motions with no applied torques.

(iii) **Arc length** from Chapter 2, computed with this metric, naturally weights each DOF by its effective inertia.

Example 3.3. Inertia-Weighted Distance in the Swing

Consider two configurations that differ by either:

(a) A $10°$ rotation of the torso (high inertia, $I \approx 12\ \mathrm{kg \cdot m^2}$), or

(b) A $10°$ rotation of the wrist (low inertia, $I \approx 0.03\ \mathrm{kg \cdot m^2}$).

In Euclidean coordinates, both represent “$10°$ of change.” In the kinetic energy metric, the torso rotation is $\sqrt{12/0.03} = 20$ times “farther” than the wrist rotation. This

correctly reflects the physical reality: rotating the torso requires 400 times more energy than rotating the wrist by the same angle.
The curvature of a trajectory measured in this metric therefore naturally captures the *physical* difficulty of the motion, not merely the angular change.

3.6.1 Geodesics as Natural Trajectories

Definition 3.8. Geodesic

A **geodesic** on a Riemannian manifold $(\mathcal{M}, \boldsymbol{G})$ is a curve that locally minimizes arc length. Geodesics satisfy the **geodesic equation**:

$$\ddot{q}^i + \sum_{j,k} \Gamma^i_{jk}(\boldsymbol{q})\, \dot{q}^j\, \dot{q}^k = 0, \tag{3.11}$$

where Γ^i_{jk} are the **Christoffel symbols** of the metric $\boldsymbol{G}$.

The remarkable fact is that the geodesic equation (3.11) is *exactly* the equation of motion for the unforced mechanical system $\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} = \boldsymbol{0}$. The Coriolis and centrifugal terms are precisely the Christoffel symbol terms. Free motions of mechanical systems are geodesics on the configuration manifold equipped with the kinetic energy metric.

Principle 3.3. Geodesics and Passive Dynamics

For trajectory design, geodesics represent the "easiest" paths—the motions the system would naturally follow with no applied forces. A well-designed trajectory stays close to geodesics where possible, deviating only where task requirements (e.g., impact conditions) demand it.
In the golf swing, the follow-through is nearly geodesic: after impact, the golfer relaxes and the body follows its natural, inertia-weighted path through configuration space. The downswing deviates significantly from the geodesic because the task (delivering maximum clubhead speed at impact) requires active torques to redirect the motion.

3.7 Curves on Manifolds Revisited

With the manifold framework in place, we can extend every concept from Chapter 2 to non-Euclidean configuration spaces.

3.7.1 Covariant Differentiation

On $\mathbb{R}^n$, we differentiate vector fields by taking component-wise derivatives. On a manifold, this does not work—the tangent spaces at different points are different vector spaces, so we cannot simply subtract vectors at different points.

Definition 3.9. Covariant Derivative

The **covariant derivative** (or **Levi-Civita connection**) ∇ is the unique way to differentiate vector fields along curves on a Riemannian manifold that:

(i) Is compatible with the metric: $\frac{\mathrm{d}}{\mathrm{d}t}\langle \boldsymbol{V}, \boldsymbol{W}\rangle = \langle \nabla_{\dot\gamma}\boldsymbol{V}, \boldsymbol{W}\rangle + \langle \boldsymbol{V}, \nabla_{\dot\gamma}\boldsymbol{W}\rangle$.

(ii) Is torsion-free: $\nabla_{\boldsymbol{V}}\boldsymbol{W} - \nabla_{\boldsymbol{W}}\boldsymbol{V} = [\boldsymbol{V}, \boldsymbol{W}]$.

In local coordinates, the covariant derivative of a vector field $\boldsymbol{V}$ along a curve $\gamma(t)$ is

$$(\nabla_{\dot\gamma}\boldsymbol{V})^i = \dot V^i + \sum_{j,k}\Gamma^i_{jk}\,\dot\gamma^j\,V^k. \tag{3.12}$$

The covariant derivative replaces the ordinary derivative in all of our Chapter 2 formulas. The generalized curvatures, the Frenet–Serret frame, and the transverse decomposition all carry over, with $\mathrm{d}/\mathrm{d}s$ replaced by $\nabla_{\dot\gamma}$.

3.7.2 Manifold Curvature of Trajectories

Definition 3.10. Geodesic Curvature

The **geodesic curvature** of a curve $\gamma(s)$ on a Riemannian manifold, parameterized by arc length, is

$$\kappa_g(s) = \|\nabla_{\dot\gamma}\dot\gamma\|\,. \tag{3.13}$$

This measures how much the curve deviates from a geodesic. A geodesic has $\kappa_g = 0$ everywhere.

Geodesic curvature is the manifold generalization of the curvature from Chapter 2, and it has the same control interpretation: *geodesic curvature measures the force per unit mass needed to follow the curve*, beyond what the natural dynamics would provide for free.

Proposition 3.1. *The control force required to track a trajectory $\gamma^*(s)$ at speed v on a configuration manifold with metric $\boldsymbol{M}(\boldsymbol{q})$ has magnitude*

$$\|\boldsymbol{\tau}_\perp\| = \kappa_g(s)\,v^2, \tag{3.14}$$

where $\|\cdot\|$ is measured in the dual (force) metric. Comparing with the Euclidean formula $a_\perp = \kappa v^2$ from Chapter 2: the Euclidean curvature κ is replaced by the geodesic curvature κ_g, which accounts for the curvature of the manifold itself.

This is a deeper version of the Curvature–Authority Principle. The control demand is reduced along directions where the manifold's own curvature "helps"—i.e., where the natural dynamics carry the system in the desired direction. The control demand increases where the trajectory fights the natural dynamics.

Example 3.4. Geodesic Curvature of the Downswing

During the downswing, the trajectory has two sources of curvature:

(a) **Euclidean curvature**: the path bends in coordinate space (the club changes direction).

(b) **Manifold curvature**: the Christoffel symbol terms ($\Gamma^i_{jk}\dot{q}^j\dot{q}^k$) represent Coriolis and centrifugal effects that curve the natural trajectories.

The geodesic curvature κ_g is the *difference*. When the Coriolis/centrifugal terms push the system in the direction the trajectory wants to go (as happens during the wrist release, where centrifugal acceleration "unhinges" the club), κ_g is *less* than the Euclidean κ. The passive dynamics are doing work for free.
This is the geometric explanation for the "free speed" of the golf swing: a well-designed trajectory aligns with the manifold's geodesics during the speed-critical phase, exploiting the natural dynamics rather than fighting them.

3.8 Error Metrics on Manifolds

We now address the critical question: how do we measure tracking error when the state space is a manifold?

3.8.1 The Logarithmic Error

Definition 3.11. Configuration Error on a Lie Group

For two configurations $\boldsymbol{q}_1, \boldsymbol{q}_2 \in \mathcal{Q}$ where $\mathcal{Q}$ is a Lie group, the **error** is

$$\boldsymbol{e} = \log(\boldsymbol{q}_1^{-1}\,\boldsymbol{q}_2) \in \text{Lie algebra}. \tag{3.15}$$

For the rotation group SO(3), this gives

$$\boldsymbol{e}_R = \log(\boldsymbol{R}_1^\top \boldsymbol{R}_2)^\vee \in \mathbb{R}^3, \tag{3.16}$$

which is the rotation vector needed to go from $\boldsymbol{R}_1$ to $\boldsymbol{R}_2$. This error is zero if and only if $\boldsymbol{q}_1 = \boldsymbol{q}_2$, and it varies smoothly with both arguments (away from the cut locus at $\|\boldsymbol{e}\| = \pi$).

Remark 3.1. Why Not Just Subtract Quaternions?

Given two unit quaternions $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$, a tempting error metric is $\boldsymbol{e} = \boldsymbol{q}_2 - \boldsymbol{q}_1$. But this lives in $\mathbb{R}^4$, not in the tangent space of S^3. Worse, it violates the double-cover property: $\boldsymbol{q}$ and $-\boldsymbol{q}$ represent the same rotation, so the error $\boldsymbol{q}_2 - \boldsymbol{q}_1$ and $-\boldsymbol{q}_2 - \boldsymbol{q}_1$ should be "the same," but they point in opposite directions.

The logarithmic error correctly handles the group structure and produces an error vector in the Lie algebra that has a clear physical interpretation (the angular velocity needed to correct the error in unit time).

3.8.2 The Trajectory Tube on a Manifold

The trajectory tube from Chapter 1 now generalizes naturally:

Definition 3.12. Manifold Trajectory Tube

Let $\boldsymbol{\gamma}^* : [0, T] \to \mathcal{Q}$ be a reference trajectory on a Riemannian manifold $(\mathcal{Q}, \boldsymbol{G})$ and let $\varepsilon : [0, T] \to \mathbb{R}_{>0}$ be a tube radius function. The **manifold trajectory tube** is

$$\mathcal{T}(t) = \{\, \boldsymbol{q} \in \mathcal{Q} : d_{\boldsymbol{G}}(\boldsymbol{q}, \boldsymbol{\gamma}^*(t)) \leq \varepsilon(t) \,\}, \tag{3.17}$$

where $d_{\boldsymbol{G}}$ is the **geodesic distance**—the length of the shortest geodesic connecting $\boldsymbol{q}$ and $\boldsymbol{\gamma}^*(t)$.

The geodesic distance correctly accounts for the topology of the configuration space. On SO(3), the maximum distance between any two orientations is π (a half-rotation), and the tube wraps properly around the group.

3.9 Equations of Motion on Manifolds

The Euler–Lagrange equations on a manifold take a coordinate-free form that reveals the geometric structure:

Theorem 3.1 (Lagrangian Mechanics on Manifolds)**.** *For a mechanical system on a configuration manifold $(\mathcal{Q}, \boldsymbol{M})$ with potential energy $V : \mathcal{Q} \to \mathbb{R}$ and generalized forces $\boldsymbol{\tau}$, the equations of motion are*

$$\boldsymbol{M}(\boldsymbol{q})\, \nabla_{\dot{\boldsymbol{q}}} \dot{\boldsymbol{q}} = -\operatorname{grad} V + \boldsymbol{B}\, \boldsymbol{u}, \tag{3.18}$$

where $\nabla_{\dot{\boldsymbol{q}}} \dot{\boldsymbol{q}}$ is the covariant acceleration and $\operatorname{grad} V$ is the Riemannian gradient of the potential energy.

In local coordinates, this reduces to the familiar $\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{C}\dot{\boldsymbol{q}} + \boldsymbol{g} = \boldsymbol{B}\boldsymbol{u}$, but the coordinate-free form (3.18) makes clear that:

- The Coriolis matrix $\boldsymbol{C}$ is not a separate "term"—it is the Christoffel symbol contribution to the covariant derivative.
- The equation has the same form on any manifold, regardless of coordinate choice.
- The "centrifugal and Coriolis forces" are geometric artifacts of the manifold's curvature, not real forces. On a flat manifold ($\mathbb{R}^n$), they vanish.

3.10 Practical Implications for Controller Implementation

We conclude with concrete guidance for implementing trajectory controllers on manifold-valued configuration spaces.

Principle 3.4. Manifold-Aware Controller Design

A trajectory tracking controller for a system on $\mathcal{Q}$ must:

(i) **Compute errors using the logarithmic map**, not coordinate subtraction:

$$\boldsymbol{e}_q = \log\big((\boldsymbol{q}^*)^{-1}\,\boldsymbol{q}\big)\,, \qquad \boldsymbol{e}_v = \dot{\boldsymbol{q}} - \mathrm{PT}_{q^*\to q}(\dot{\boldsymbol{q}}^*), \tag{3.19}$$

where PT denotes parallel transport of the reference velocity to the current configuration.

(ii) **Compute feedforward using the covariant derivative**, not ordinary differentiation of coordinates.

(iii) **Design feedback gains in the tangent space**, then map the resulting control back to the manifold via the exponential map.

(iv) **Monitor for cut-locus proximity**: when the error approaches π (for SO(3)), the logarithmic map becomes ill-conditioned. Switch to a recovery strategy.

Example 3.5. PD Control on SO(3)

The manifold-correct PD controller for tracking a reference orientation $\boldsymbol{R}^*(t)$ with a rigid body (moment of inertia $\boldsymbol{J}$, angular velocity $\boldsymbol{\omega}$) is:

$$\boldsymbol{\tau} = -k_p\,\log(\boldsymbol{R}^{*\top}\boldsymbol{R})^\vee - k_d\,(\boldsymbol{\omega} - \boldsymbol{R}^{*\top}\boldsymbol{R}\,\boldsymbol{\omega}^*) + \boldsymbol{J}\dot{\boldsymbol{\omega}}^* + \boldsymbol{\omega}\times\boldsymbol{J}\boldsymbol{\omega}, \tag{3.20}$$

where the first term is the manifold-correct proportional error (logarithmic map), the second is the velocity error (parallel-transported), and the last two terms are the feedforward (covariant derivative of the reference).

Compare with the naïve Euler-angle PD controller: $\tau = -k_p(\boldsymbol{\theta} - \boldsymbol{\theta}^*) - k_d(\dot{\boldsymbol{\theta}} - \dot{\boldsymbol{\theta}}^*)$. The naïve controller has singularities; Eq. (3.20) does not. The naïve controller produces torques proportional to angle differences (which can be misleading near $\pm\pi$); Eq. (3.20) produces torques proportional to the *true rotation* needed to correct the error.

3.11 Summary

Euclidean assumption	**Manifold replacement**
$\mathcal{Q} = \mathbb{R}^n$	$\mathcal{Q} = \mathrm{SO}(3)^k \times \mathbb{T}^m \times \cdots$
Error: $\boldsymbol{e} = \boldsymbol{q} - \boldsymbol{q}^*$	Error: $\boldsymbol{e} = \log((\boldsymbol{q}^*)^{-1}\boldsymbol{q})$
Derivative: $\ddot{\boldsymbol{q}}$	Covariant derivative: $\nabla_{\dot{q}}\dot{\boldsymbol{q}}$
Shortest path: straight line	Shortest path: geodesic
Distance: $\|\boldsymbol{q}_2 - \boldsymbol{q}_1\|$	Geodesic distance: $d_{\boldsymbol{G}}(\boldsymbol{q}_1, \boldsymbol{q}_2)$
Curvature: κ (Ch. 2)	Geodesic curvature: κ_g
Trajectory tube: Euclidean ball	Geodesic ball
Coriolis terms: "extra forces"	Christoffel symbols: manifold geometry
PD: $-k_p\boldsymbol{e} - k_d\dot{\boldsymbol{e}}$	$-k_p \log(\cdot)^\vee - k_d$(parallel transport)

The central message of this chapter is that the configuration spaces of mechanical systems have geometry, and this geometry matters for control. Working in $\mathbb{R}^n$ when the true space is $\mathrm{SO}(3) \times \mathbb{T}^k$ introduces artificial singularities, incorrect error metrics, and controllers that fail when the system moves far from a nominal configuration.

The trajectory-first philosophy of this book demands that we take the geometry seriously from the start. The curves we designed in Chapter 2 live on manifolds, and the stability theory we will develop in Chapter 4 must respect the manifold structure. The payoff is controllers that work globally, without singularities, and that correctly exploit the natural dynamics encoded in the manifold's Riemannian geometry.

3.12 Exercises

3.1. **(Topology.)** Determine the configuration manifold and its dimension for each system:

(a) A planar double pendulum (two revolute joints in 2D).

(b) A spherical pendulum (a point mass on a rigid rod, free to swing in all directions).

(c) A spacecraft with two reaction wheels (the body rotates freely in 3D; each wheel adds one internal DOF).

For each, identify whether $\mathbb{R}^n$ coordinates would have singularities.

3.2. **(Euler angle singularity.)** For the ZYX Euler angle representation $\boldsymbol{R} = R_z(\phi)\, R_y(\theta)\, R_x(\psi)$:

(a) Compute the Jacobian $\boldsymbol{J}$ relating angular velocity $\boldsymbol{\omega}$ to Euler angle rates $(\dot{\phi}, \dot{\theta}, \dot{\psi})$.

(b) Show that $\boldsymbol{J}$ is singular at $\theta = \pm\pi/2$.

(c) Find an orientation trajectory $\boldsymbol{R}(t)$ that passes through the singularity and compute the Euler angle rates. What happens?

3.3. (Logarithmic error.) For two rotations $\boldsymbol{R}_1 = \boldsymbol{I}$ and $\boldsymbol{R}_2 = R_z(\alpha)$ (rotation by α about z):

(a) Compute $\log(\boldsymbol{R}_1^\top \boldsymbol{R}_2)^\vee$ for $\alpha = \pi/6, \pi/2, \pi, 5\pi/3$.

(b) Compare with the "error" $\alpha - 0 = \alpha$ as a function of α. At what point do they differ significantly?

(c) What happens at $\alpha = \pi$? Explain geometrically.

3.4. (Geodesics.) For a double pendulum with masses m_1, m_2 and lengths l_1, l_2:

(a) Write the mass matrix $\boldsymbol{M}(\theta_1, \theta_2)$.

(b) Compute the Christoffel symbols (or use the Coriolis matrix formula $C_{ijk} = \frac{1}{2}(\partial M_{ij}/\partial q_k + \partial M_{ik}/\partial q_j - \partial M_{jk}/\partial q_i)$).

(c) Numerically integrate the geodesic equation from initial condition $(\theta_1, \theta_2) = (0, 0)$ with $(\dot\theta_1, \dot\theta_2) = (1, 0)$. Plot the geodesic on $\mathbb{T}^2$.

3.5. (Manifold PD control.) Implement and compare:

(a) An Euler-angle PD controller for a rigid body tracking a reference orientation $\boldsymbol{R}^*(t) = R_z(\omega t)$ (constant-speed rotation).

(b) The manifold PD controller from Eq. (3.20) for the same task.

(c) Run both controllers with the reference trajectory passing through $\theta = \pi/2$ (gimbal lock for ZYX convention). Compare the tracking errors and commanded torques.

3.6. (Golf swing configuration space.) For the 9-DOF golf swing model of Eq. (3.8):

(a) How many Euler angles would be needed to parameterize the full configuration space? How many singularity surfaces exist?

(b) Propose a quaternion-based parameterization for the two SO(3) joints. What is the total number of parameters? What constraints must be maintained?

(c) Discuss the trade-offs between the Euler angle and quaternion representations for numerical simulation of the golf swing.

The sphere cannot be flattened without tearing.
The rotation group cannot be charted without singularity.

Do not fight the topology—ride it.

The manifold is not an obstacle. It is the landscape through which every motion flows.

Chapter 4

Orbital Stability and Transverse Linearization

Stability is not the absence of motion.
It is the faithfulness of motion.

4.1 The Problem With Lyapunov

The classical Lyapunov framework asks a simple question: *does the state converge to an equilibrium?* We demonstrated in Chapter 1 that this is the wrong question for systems whose purpose is to move. But the problem goes deeper than philosophical framing—Lyapunov stability is *structurally incapable* of analyzing trajectory-tracking systems.

Consider a system perfectly tracking a reference trajectory $\boldsymbol{\gamma}^*(t)$. The state $\boldsymbol{x}(t)$ matches $\boldsymbol{\gamma}^*(t)$ at every instant. Now apply a small timing perturbation: the system executes the same motion but shifted by ε seconds. The perturbed state is $\tilde{\boldsymbol{x}}(t) = \boldsymbol{\gamma}^*(t - \varepsilon)$, and the time-indexed error is

$$\boldsymbol{e}(t) = \tilde{\boldsymbol{x}}(t) - \boldsymbol{\gamma}^*(t) = \boldsymbol{\gamma}^*(t - \varepsilon) - \boldsymbol{\gamma}^*(t) \approx -\varepsilon\, \dot{\boldsymbol{\gamma}}^*(t). \tag{4.1}$$

This error does *not* decay to zero—it persists indefinitely, oscillating with a magnitude proportional to $\varepsilon\, v(t)$ where v is the speed. By Lyapunov's definition, the trajectory is *unstable*. But the system is executing the correct motion perfectly! The only "error" is that it is slightly ahead of or behind schedule.

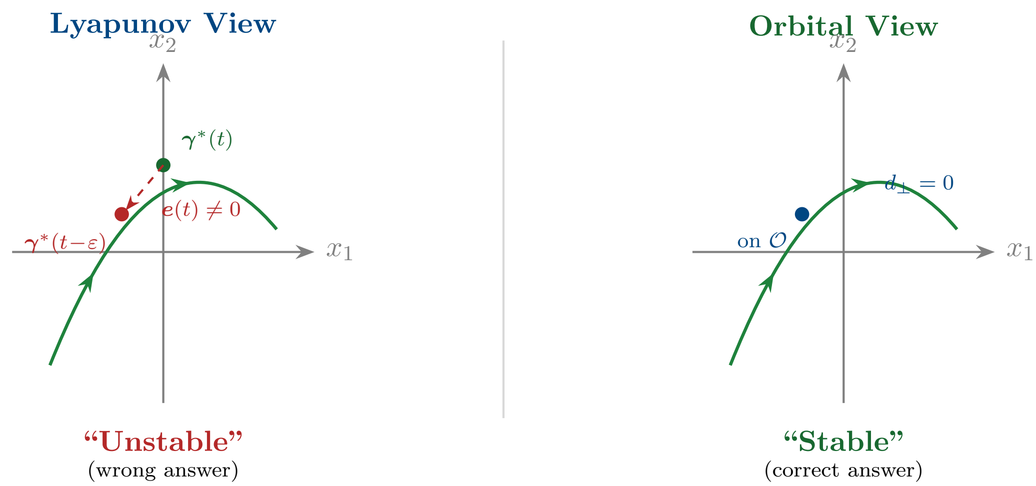
Key Idea 4.1. The Fundamental Inadequacy of Point Stability

Lyapunov stability conflates two independent phenomena:

(i) **Transverse deviations**: the system drifts *away from* the trajectory path. This is a genuine control failure.

(ii) **Tangential deviations**: the system is ahead of or behind schedule *along* the trajectory path. This is a timing difference, not a tracking failure.

Lyapunov stability rejects both. Orbital stability accepts tangential deviations and rejects only transverse ones. For systems whose purpose is motion, orbital stability is the correct notion.



4.2 Orbital Stability: Formal Definitions

We now define orbital stability rigorously. The key object is the **orbit**—the image of the trajectory as a set, stripped of all timing information.

Definition 4.1. Orbit

Given a trajectory $\boldsymbol{\gamma}^* : [0, T] \to \mathcal{X}$, the **orbit** is the set

$$\mathcal{O} = \{\boldsymbol{\gamma}^*(t) : t \in [0, T]\} \subset \mathcal{X}. \tag{4.2}$$

The orbit is a one-dimensional submanifold of $\mathcal{X}$ (a curve without parameterization). For periodic trajectories with period T, $\boldsymbol{\gamma}^*(0) = \boldsymbol{\gamma}^*(T)$ and $\mathcal{O}$ is a closed curve (a loop).

Definition 4.2. Orbital Stability

A trajectory $\boldsymbol{\gamma}^*(t)$ (or its orbit $\mathcal{O}$) is **orbitally stable** if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d(\boldsymbol{x}(0), \mathcal{O}) < \delta \implies d(\boldsymbol{x}(t), \mathcal{O}) < \varepsilon \quad \forall t \geq 0, \tag{4.3}$$

where $d(\boldsymbol{x}, \mathcal{O}) = \inf_{p \in \mathcal{O}} \|\boldsymbol{x} - p\|$ is the distance from the state to the orbit.
It is **asymptotically orbitally stable** if additionally $d(\boldsymbol{x}(t), \mathcal{O}) \to 0$ as $t \to \infty$.

Critically, orbital stability does not require $\|\boldsymbol{x}(t) - \boldsymbol{\gamma}^*(t)\| \to 0$. The system may drift along the orbit (ahead or behind schedule) while remaining perfectly stable. Only deviations *away from* the orbit matter.

Definition 4.3. Orbital Stability with Asymptotic Phase

A trajectory is orbitally stable **with asymptotic phase** if it is asymptotically orbitally stable and additionally there exists a phase shift θ^* such that

$$\|\boldsymbol{x}(t) - \boldsymbol{\gamma}^*(t + \theta^*)\| \to 0 \quad \text{as } t \to \infty. \tag{4.4}$$

This means the system not only returns to the orbit but eventually "locks in" to a definite timing, shifted by θ^* from the nominal schedule. This is the strongest practical form of trajectory stability.

Example 4.1. Limit Cycles: The Prototype

The Van der Pol oscillator $\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$ possesses a **limit cycle**—a periodic orbit that is asymptotically orbitally stable with asymptotic phase. All nearby trajectories spiral toward the limit cycle and eventually synchronize to a definite phase. The limit cycle is the simplest example of a system whose purpose is to move: the "target" is not a point but an orbit, and stability means the orbit attracts nearby trajectories. Walking, running, and heartbeats are all limit cycles. The golf swing is a transient trajectory (not periodic), but the same stability concepts apply over the finite interval $[0, T]$.

4.3 The Transverse–Tangential Decomposition

To analyze orbital stability, we must decompose the dynamics into components along and across the orbit. This is the *transverse–tangential decomposition*, the central technical tool of this chapter.

4.3.1 Projection Onto the Orbit

Definition 4.4. Nearest-Point Projection

Given a state $\boldsymbol{x}$ sufficiently close to the orbit $\mathcal{O}$, the **nearest-point projection** is

$$\pi(\boldsymbol{x}) = \arg\min_{p \in \mathcal{O}} \|\boldsymbol{x} - p\|, \tag{4.5}$$

and the associated **projected phase** is the unique $s^*(\boldsymbol{x})$ satisfying $\pi(\boldsymbol{x}) = \boldsymbol{\gamma}^*(s^*)$. The projection is well-defined and smooth in a tubular neighborhood of $\mathcal{O}$ (as long as $\boldsymbol{x}$ is closer to $\mathcal{O}$ than the minimum radius of curvature of the orbit).

4.3.2 The Moving Hyperplane Decomposition

At each point $\boldsymbol{\gamma}^*(s)$ on the orbit, the state space $\mathbb{R}^n$ decomposes into a one-dimensional tangential direction and an $(n-1)$-dimensional transverse hyperplane:

Definition 4.5. Transverse Hyperplane

At the point $\boldsymbol{\gamma}^*(s)$ on the orbit, define:

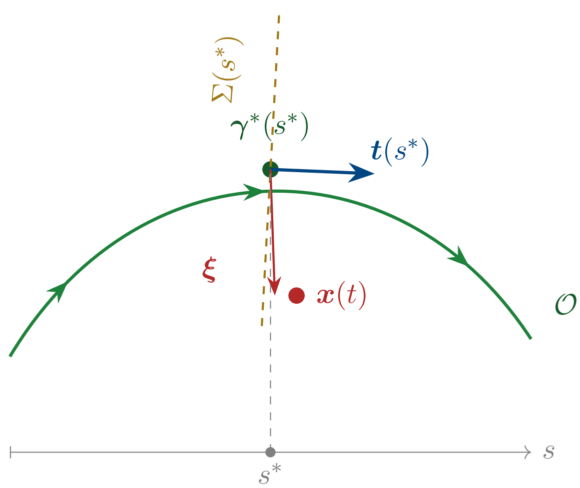
$$\text{Tangential direction:} \quad \boldsymbol{t}(s) = \frac{\dot{\boldsymbol{\gamma}}^*(s)}{\|\dot{\boldsymbol{\gamma}}^*(s)\|}, \tag{4.6}$$

$$\text{Transverse hyperplane:} \quad \Sigma(s) = \{\boldsymbol{v} \in \mathbb{R}^n : \boldsymbol{v}^\top \boldsymbol{t}(s) = 0\}. \tag{4.7}$$

A state $\boldsymbol{x}$ near $\boldsymbol{\gamma}^*(s)$ decomposes as

$$\boldsymbol{x} = \boldsymbol{\gamma}^*(s^*) + \boldsymbol{\xi}, \qquad \boldsymbol{\xi} \in \Sigma(s^*), \tag{4.8}$$

where s^* is the projected phase and $\boldsymbol{\xi}$ is the **transverse deviation**—the component of the error perpendicular to the orbit. Orbital stability is equivalent to the stability of the origin $\boldsymbol{\xi} = \mathbf{0}$ in the transverse dynamics.



4.4 Transverse Linearization

The power of the transverse decomposition is that it reduces orbital stability to the stability of a *time-varying linear system*—the transverse linearization.

4.4.1 Deriving the Transverse Dynamics

Consider the nonlinear system $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})$ with a reference trajectory $\boldsymbol{\gamma}^*(t)$ satisfying $\dot{\boldsymbol{\gamma}}^* = \boldsymbol{f}(\boldsymbol{\gamma}^*, \boldsymbol{u}^*(t))$. Using the decomposition $\boldsymbol{x} = \boldsymbol{\gamma}^*(s^*) + \boldsymbol{\xi}$ and linearizing about $\boldsymbol{\xi} = \mathbf{0}$:

Theorem 4.1 (Transverse Linearization)**.** *The linearized transverse dynamics about the orbit $\mathcal{O}$ are*

$$\dot{\boldsymbol{\xi}} = \boldsymbol{A}_\perp(s)\, \boldsymbol{\xi} + \boldsymbol{B}_\perp(s)\, \delta \boldsymbol{u}, \tag{4.9}$$

where $\delta\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{u}^*$ is the feedback perturbation, and

$$\boldsymbol{A}_\perp(s) = \boldsymbol{P}(s)\left[\left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{\boldsymbol{\gamma}^*,\boldsymbol{u}^*} - \frac{\dot{\boldsymbol{\gamma}}^* \otimes (\nabla_{\boldsymbol{x}} s^*)^\top}{\|\dot{\boldsymbol{\gamma}}^*\|^2}\cdot\left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{\boldsymbol{\gamma}^*,\boldsymbol{u}^*}\right]\boldsymbol{P}(s), \tag{4.10}$$

$$\boldsymbol{B}_\perp(s) = \boldsymbol{P}(s)\left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right|_{\boldsymbol{\gamma}^*,\boldsymbol{u}^*}, \tag{4.11}$$

and $\boldsymbol{P}(s) = \boldsymbol{I} - \boldsymbol{t}(s)\boldsymbol{t}(s)^\top$ is the orthogonal projector onto $\Sigma(s)$.

Remark 4.1. Understanding the Transverse $\boldsymbol{A}_\perp$

Equation (4.10) has a clear structure:

- The first term $\boldsymbol{P}\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\boldsymbol{P}$ is the standard Jacobian projected onto the transverse space. This captures how the system dynamics affect cross-trajectory deviations.
- The second term subtracts the component of the Jacobian that acts along the tangent direction—this is exactly the tangential dynamics that orbital stability "forgets."

The resulting $\boldsymbol{A}_\perp(s)$ has dimension $(n-1) \times (n-1)$, one less than the full state Jacobian. The removed eigenvalue corresponds to motion along the orbit—the tangential direction in which we are *not* demanding stability.

Principle 4.1. The Transverse Reduction Principle

The reference trajectory $\boldsymbol{\gamma}^*(t)$ is asymptotically orbitally stable if and only if the origin of the transverse linearization (4.9) (with $\delta\boldsymbol{u} = \boldsymbol{0}$) is asymptotically stable. This reduces the orbital stability question from a nonlinear problem on the full n-dimensional state space to a *linear* problem on the $(n-1)$-dimensional transverse space. For controller design: if we can stabilize the linear system (4.9), we achieve orbital stability of the nonlinear system. All the tools of linear time-varying (LTV) control theory apply.

4.4.2 The Transverse Jacobian: A Practical Formula

For implementation, the key quantities are computed as follows. Let $\boldsymbol{J}(t) = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{\boldsymbol{\gamma}^*,\boldsymbol{u}^*}$ be the Jacobian along the reference. Choose any orthonormal basis $\{\boldsymbol{n}_1(s), \ldots, \boldsymbol{n}_{n-1}(s)\}$ for $\Sigma(s)$—the Frenet–Serret frame from Chapter 2 provides a natural choice. Collect these into the matrix $\boldsymbol{N}(s) = [\boldsymbol{n}_1 \mid \cdots \mid \boldsymbol{n}_{n-1}] \in \mathbb{R}^{n\times(n-1)}$. Then:

$$\boldsymbol{A}_\perp(s) = \boldsymbol{N}(s)^\top\left[\boldsymbol{J}(s) - \frac{\boldsymbol{f}^*\cdot\boldsymbol{J}(s)^\top\boldsymbol{t}(s)}{\|\boldsymbol{f}^*\|^2}\,\boldsymbol{f}^*\,\boldsymbol{t}(s)^\top - \dot{\boldsymbol{N}}(s)\,\boldsymbol{N}(s)^\top\right]\boldsymbol{N}(s), \tag{4.12}$$

where $\boldsymbol{f}^* = \boldsymbol{f}(\boldsymbol{\gamma}^*, \boldsymbol{u}^*)$ and $\dot{\boldsymbol{N}}$ accounts for the rotation of the transverse frame along the orbit (the Frenet–Serret equations).

Example 4.2. Transverse Linearization of a Planar System

Consider the system $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1 + u$ tracking the circular orbit $\boldsymbol{\gamma}^*(t) = (\cos t, -\sin t)$ with $u^* = 0$.
The Jacobian is $\boldsymbol{J} = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, the tangent vector is $\boldsymbol{t} = (-\sin t, -\cos t)^\top/1 = (-\sin t, -\cos t)^\top$, and the normal is $\boldsymbol{n} = (-\cos t, \sin t)^\top$.
Computing $\boldsymbol{A}_\perp = \boldsymbol{n}^\top[\boldsymbol{J} - \text{tangential terms}]\boldsymbol{n}$, we find:

$$\boldsymbol{A}_\perp(t) = 0, \qquad \boldsymbol{B}_\perp(t) = \boldsymbol{n}^\top \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \sin t. \tag{4.13}$$

The open-loop transverse dynamics are *neutrally stable* ($\boldsymbol{A}_\perp = 0$). Feedback $\delta u = -k\,\xi$ gives $\dot{\xi} = -k\sin^2 t\,\xi$ (after averaging), which is stable for $k > 0$. The orbit is stabilizable.

4.5 Floquet Theory: Stability of Periodic Orbits

For periodic trajectories (limit cycles, walking gaits, repetitive manufacturing), the transverse linearization is a *periodic* linear system. Floquet theory provides the definitive stability analysis tool.

4.5.1 The State Transition Matrix

Definition 4.6. State Transition Matrix

For the linear time-varying system $\dot{\boldsymbol{\xi}} = \boldsymbol{A}_\perp(t)\,\boldsymbol{\xi}$, the **state transition matrix** $\boldsymbol{\Phi}(t, t_0)$ satisfies

$$\dot{\Phi}(t, t_0) = \boldsymbol{A}_\perp(t)\,\boldsymbol{\Phi}(t, t_0), \qquad \boldsymbol{\Phi}(t_0, t_0) = \boldsymbol{I}. \tag{4.14}$$

The solution is $\boldsymbol{\xi}(t) = \boldsymbol{\Phi}(t, t_0)\,\boldsymbol{\xi}(t_0)$.

4.5.2 The Monodromy Matrix

Definition 4.7. Monodromy Matrix

For a periodic orbit with period T, the **monodromy matrix** is the state transition matrix over one complete period:

$$\boldsymbol{M} = \boldsymbol{\Phi}(T, 0). \tag{4.15}$$

The eigenvalues $\mu_1, \ldots, \mu_{n-1}$ of $\boldsymbol{M}$ are the **Floquet multipliers** (also called **characteristic multipliers**).

Theorem 4.2 (Floquet Stability Criterion)**.** *A periodic orbit is:*

(i) ***Orbitally stable*** *if all Floquet multipliers satisfy* $|\mu_k| \leq 1$*, with multipliers on the unit circle being semisimple.*

(ii) ***Asymptotically orbitally stable*** *if all Floquet multipliers satisfy* $|\mu_k| < 1$ *(all strictly inside the unit circle).*

(iii) ***Orbitally unstable*** *if any Floquet multiplier satisfies* $|\mu_k| > 1$*.*

Remark 4.2. The Missing Multiplier

If we had computed the monodromy matrix for the *full* n-dimensional system (not the transverse reduction), there would be n Floquet multipliers, with one of them exactly equal to $+1$. This "trivial" multiplier corresponds to the tangential direction—perturbations along the orbit neither grow nor decay. The transverse reduction removes this trivial multiplier, leaving only the $n-1$ multipliers that matter for orbital stability.

This is why the transverse reduction is essential: without it, the monodromy matrix *always* has an eigenvalue at $+1$, making the orbit appear only marginally stable even when it is asymptotically orbitally stable.

Example 4.3. Floquet Analysis of a Walking Gait

Consider a simplified compass-gait biped with 4-dimensional state space $(\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2)$. The walking gait is a periodic orbit with period T (one stride). The transverse linearization has dimension $n - 1 = 3$.

After numerically integrating $\mathbf{\Phi}(T, 0)$ for the transverse dynamics, suppose we find the monodromy matrix has eigenvalues $\mu_1 = 0.7$, $\mu_2 = 0.4$, $\mu_3 = -0.3$. Since all $|\mu_k| < 1$, the gait is asymptotically orbitally stable.

The physical interpretation: $\mu_1 = 0.7$ means that after one stride, transverse deviations in the first mode are reduced to 70% of their initial value. After 10 strides, they are reduced to $0.7^{10} \approx 3\%$. The negative multiplier $\mu_3 = -0.3$ indicates an alternating (flip-flop) convergence pattern in the third mode.

For the golf swing (a non-periodic motion), we cannot use the monodromy matrix directly, but the state transition matrix $\mathbf{\Phi}(T, 0)$ plays an analogous role—its singular values measure how much transverse deviations amplify or attenuate over the course of the motion.

4.6 Moving Poincaré Sections

A complementary tool for analyzing orbital stability is the **Poincaré section**—a codimension-1 surface that the orbit crosses transversally. While Floquet theory gives continuous-time information, Poincaré sections give *discrete-time* (strobed) information.

Definition 4.8. Poincaré Section

A **Poincaré section** for an orbit $\mathcal{O}$ is a smooth $(n-1)$-dimensional surface $\mathcal{S} \subset \mathcal{X}$ such that:

(i) $\mathcal{O}$ intersects $\mathcal{S}$ transversally (not tangentially).

(ii) In a neighborhood of the intersection, every orbit near $\mathcal{O}$ also crosses $\mathcal{S}$.

The **Poincaré return map** $\boldsymbol{P} : \mathcal{S} \to \mathcal{S}$ sends each point on $\mathcal{S}$ to the next intersection with $\mathcal{S}$ under the flow.

For periodic orbits, the orbit crosses $\mathcal{S}$ once per period, and the Poincaré return map has a fixed point at the intersection $\boldsymbol{x}^* = \mathcal{O} \cap \mathcal{S}$. Orbital stability of $\mathcal{O}$ is equivalent to stability of this fixed point under the return map.

Theorem 4.3 (Poincaré–Floquet Connection)**.** *The Jacobian of the Poincaré return map at the fixed point equals the transverse monodromy matrix:*

$$D\boldsymbol{P}(\boldsymbol{x}^*) = \boldsymbol{M}_\perp. \tag{4.16}$$

The Floquet multipliers and the eigenvalues of the Poincaré map are identical.

4.6.1 Moving Poincaré Sections for Non-Periodic Trajectories

The golf swing is not periodic—it is a *finite-time* trajectory. The classical Poincaré section (which relies on periodicity) does not directly apply. However, we can generalize to **moving Poincaré sections**: a continuous family of transverse hyperplanes, one at each point of the orbit.

Definition 4.9. Moving Poincaré Section

A **moving Poincaré section** along a trajectory $\boldsymbol{\gamma}^*(s)$, $s \in [0, L]$, is the family of hyperplanes

$$\mathcal{S}(s) = \boldsymbol{\gamma}^*(s) + \Sigma(s), \tag{4.17}$$

where $\Sigma(s)$ is the transverse hyperplane at s. The **section-to-section map** from $\mathcal{S}(s_1)$ to $\mathcal{S}(s_2)$ is defined by following the flow from s_1 to s_2:

$$\boldsymbol{P}_{s_1 \to s_2} : \mathcal{S}(s_1) \to \mathcal{S}(s_2), \qquad \boldsymbol{\xi}(s_1) \mapsto \boldsymbol{\xi}(s_2). \tag{4.18}$$

The linearized section-to-section map is the state transition matrix $\boldsymbol{\Phi}_\perp(s_2, s_1)$ of the transverse dynamics.

For the golf swing, this gives us a powerful analysis tool:

Principle 4.2. Finite-Time Orbital Stability

For a finite-time trajectory $\boldsymbol{\gamma}^* : [0, T] \to \mathcal{X}$, define the **transverse amplification ratio** from phase s_1 to s_2:

$$\rho(s_1, s_2) = \|\boldsymbol{\Phi}_\perp(s_2, s_1)\|_2 = \sigma_{\max}(\boldsymbol{\Phi}_\perp(s_2, s_1)), \tag{4.19}$$

where $\sigma_{\max}$ is the largest singular value. If $\rho(s_1, s_2) < 1$, then transverse deviations shrink between s_1 and s_2. If $\rho(s_1, s_2) > 1$, they grow.
For trajectory design, we require:

$$\rho(0, s_{\text{impact}}) \ll 1 \qquad \text{(deviations shrink toward impact).} \tag{4.20}$$

This is a precise formulation of the funnel concept from Chapter 1: the tube narrows because the transverse dynamics are contracting.

Example 4.4. Amplification Profile of the Golf Swing

Consider the 8-dimensional state space of a 4-DOF golf swing model. The transverse dynamics have dimension 7. The transverse state transition matrix $\boldsymbol{\Phi}_\perp(s, 0)$ maps initial deviations to deviations at phase s.
Numerical computation reveals the amplification ratio $\rho(0, s)$ as a function of s:

- $s \in [0, 0.3]$ (backswing): ρ is approximately 1—deviations neither grow nor shrink. The motion is kinematically slow and nearly linear.
- $s \in [0.3, 0.5]$ (transition): ρ *increases* past 1. This is the unstable phase—the high curvature and speed change amplify deviations. Without feedback, errors grow here.
- $s \in [0.5, 0.7]$ (late downswing): ρ *decreases*. The passive dynamics of the double-pendulum release are self-stabilizing in the transverse directions. The wrist release "funnels" the club toward the correct impact state.
- $s = 0.75$ (impact): $\rho(0, 0.75) < 1$. The net effect over the entire swing is transverse contraction—deviations are smaller at impact than at address.

This profile explains why skilled golfers are accurate despite the extreme nonlinearity of the swing: the trajectory is designed so that its *passive transverse dynamics* are contracting near impact, creating a natural funnel that corrects errors without active feedback.

4.7 Controller Design via Transverse Stabilization

We now use the transverse linearization to design tracking controllers. The approach is systematic: stabilize the transverse linear system (4.9) using standard LTV techniques, then map the result back to the nonlinear system.

4.7.1 Time-Varying LQR for Transverse Stabilization

Principle 4.3. Transverse LQR

The trajectory tracking problem reduces to solving the *differential Riccati equation* for the transverse dynamics:

$$-\dot{\boldsymbol{S}}(t) = \boldsymbol{A}_\perp^\top \boldsymbol{S} + \boldsymbol{S}\,\boldsymbol{A}_\perp - \boldsymbol{S}\,\boldsymbol{B}_\perp \boldsymbol{R}^{-1} \boldsymbol{B}_\perp^\top \boldsymbol{S} + \boldsymbol{Q}(t), \tag{4.21}$$

with terminal condition $\boldsymbol{S}(T) = \boldsymbol{S}_T$. The optimal transverse feedback gain is

$$\boldsymbol{K}_\perp(t) = \boldsymbol{R}^{-1}\,\boldsymbol{B}_\perp(t)^\top\,\boldsymbol{S}(t). \tag{4.22}$$

The full control law is

$$\boldsymbol{u}(t) = \underbrace{\boldsymbol{u}^*(t)}_{\text{feedforward}} + \underbrace{\boldsymbol{N}(s^*)\,\boldsymbol{K}_\perp(t)\,\boldsymbol{N}(s^*)^\top(\boldsymbol{x} - \boldsymbol{\gamma}^*(s^*))}_{\text{transverse feedback}}, \tag{4.23}$$

where $\boldsymbol{N}(s^*)$ maps between the transverse coordinates $\boldsymbol{\xi}$ and the full state coordinates.

The weighting matrices $\boldsymbol{Q}(t)$ and $\boldsymbol{R}$ have clear physical interpretations:

- $\boldsymbol{Q}(t)$ penalizes transverse deviation. Making $\boldsymbol{Q}$ large near impact tightens the funnel where precision matters.
- $\boldsymbol{R}$ penalizes control effort. Making $\boldsymbol{R}$ small allows aggressive correction but may demand unrealistic actuator authority.
- The *phase-dependent* nature of $\boldsymbol{Q}(t)$ is essential: we want tight tracking near impact and loose tracking during the backswing. This is the time-varying tube from Chapter 1, now realized through the Riccati equation.

4.7.2 Phase-Varying Gains and the Funnel Connection

The terminal-value Riccati equation (4.21) naturally produces gains that tighten as the terminal time approaches. This is exactly the funnel behavior we designed geometrically in Chapter 1:

Theorem 4.4 (Riccati Funnel)**.** *Let $\boldsymbol{S}(t)$ be the solution of (4.21) with terminal penalty $\boldsymbol{S}_T \succ 0$. The ellipsoidal set*

$$\mathcal{E}(t) = \left\{\boldsymbol{\xi} \in \Sigma(s) : \boldsymbol{\xi}^\top \boldsymbol{S}(t)\,\boldsymbol{\xi} \le c\right\} \tag{4.24}$$

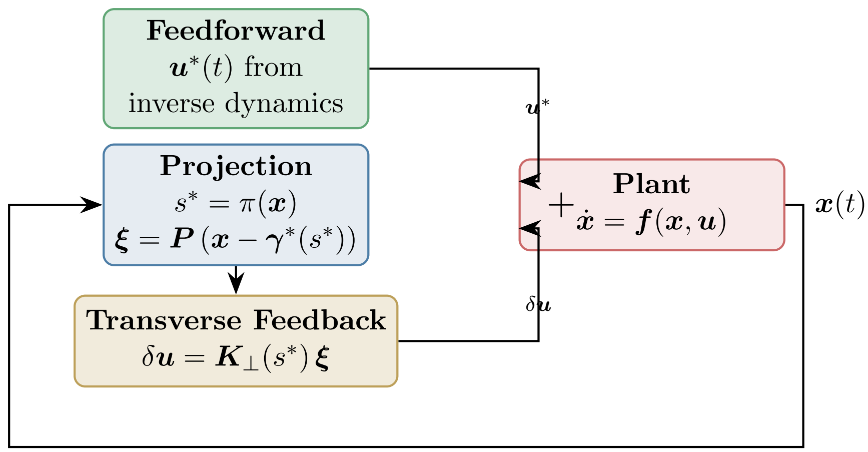
defines a time-varying tube whose width (in the direction of eigenvector $\boldsymbol{v}_k$ of $\boldsymbol{S}$) is proportional to $1/\sqrt{\lambda_k(t)}$, where $\lambda_k(t)$ is the corresponding eigenvalue of $\boldsymbol{S}(t)$.

As $t \to T$ and $\boldsymbol{S}(t) \to \boldsymbol{S}_T$, the tube tightens. The closed-loop transverse dynamics ensure that any trajectory starting inside the tube at $t = 0$ remains inside for all $t \in [0, T]$, provided the control authority is sufficient.

This is the formal connection between the geometric funnel of Chapter 1 and the algebraic Riccati equation: **the funnel is the level set of the Riccati cost-to-go.**

4.7.3 The Complete Tracking Controller Architecture

Assembling all the pieces, the trajectory tracking controller has three layers:



Layer 1. **Feedforward $\boldsymbol{u}^*(t)$**: computed offline from inverse dynamics, provides ∼90% of the total control effort.

Layer 2. **Projection**: measures the current state $\boldsymbol{x}(t)$, projects onto the orbit to find s^*, and computes the transverse deviation $\boldsymbol{\xi}$.

Layer 3. **Transverse feedback $\delta\boldsymbol{u} = \boldsymbol{K}_\perp(s^*)\,\boldsymbol{\xi}$**: stabilizes the transverse dynamics using phase-varying gains from the Riccati equation.

Remark 4.3. Why Not Just Use Standard LQR?

One could linearize the *full* dynamics about $\boldsymbol{\gamma}^*(t)$ and apply standard time-varying LQR to the n-dimensional error $\boldsymbol{e}(t) = \boldsymbol{x}(t) - \boldsymbol{\gamma}^*(t)$. This works, but:

(a) It "wastes" control effort trying to correct tangential deviations (timing errors), which are benign.

(b) It fights the inherent marginal stability in the tangential direction, leading to unnecessarily high gains.

(c) It does not produce the correct funnel shape—it penalizes all errors equally, including timing errors.

The transverse approach avoids all three problems. It is geometrically correct, computationally smaller (dimension $n - 1$ vs. n), and physically meaningful.

4.8 Robustness of Orbital Stability

A practical controller must be robust to model uncertainty and external disturbances. The transverse framework provides natural robustness margins.

Definition 4.10. Transverse Gain and Phase Margins

For the closed-loop transverse system $\dot{\boldsymbol{\xi}} = [\boldsymbol{A}_\perp - \boldsymbol{B}_\perp \boldsymbol{K}_\perp]\,\boldsymbol{\xi}$, the **transverse gain margin** and **transverse phase margin** are the gain and phase margins of the transverse loop transfer function. These are the trajectory-tracking analogues of the classical setpoint stability margins.

Proposition 4.1 (Robustness of Transverse LQR)**.** *If the transverse LQR is designed with $\boldsymbol{R} = r\boldsymbol{I}$, then the closed-loop transverse system has guaranteed gain margin $[1/2, \infty)$ and phase margin $\pm 60°$, inheriting the classical LQR robustness guarantees in each transverse channel.*

For the golf swing, robustness translates directly to *repeatability*: a swing with high transverse robustness margins produces consistent results even when the golfer's motor commands are noisy.

4.9 Summary

Concept	Significance
Orbital stability	The correct stability notion for moving systems: only cross-orbit deviations matter
Transverse linearization	Reduces orbital stability to $(n-1)$-dim LTV stability; removes the tangential "nuisance" direction
Floquet multipliers	Eigenvalues of the transverse monodromy matrix; determine stability of periodic orbits
Moving Poincaré sections	Extend Poincaré analysis to non-periodic trajectories; connect to the state transition matrix
Amplification ratio ρ	Maximum singular value of $\boldsymbol{\Phi}_\perp$; quantifies transverse error growth or contraction
Transverse LQR	Phase-varying feedback from the Riccati equation; the optimal funnel controller
Riccati funnel	The LQR cost-to-go defines the funnel boundary; connects geometric and algebraic views

This chapter has provided the mathematical backbone for everything that follows. The transverse linearization converts the nonlinear trajectory-tracking problem into a linear, lower-dimensional problem that inherits all the powerful tools of LTV control theory. The Floquet/Poincaré framework provides both analytical insight and numerical algorithms for assessing stability.

The key insight for the golf swing is that orbital stability—not Lyapunov stability—is the correct framework. The swing is designed so that its passive transverse dynamics create a natural funnel toward impact, and the transverse LQR controller tightens this funnel using phase-varying gains that match the demands of each phase of the motion.

In Chapter 5, we will see how underactuation constrains the transverse dynamics, creating geometric structure that can be *exploited* rather than merely tolerated.

4.10 Exercises

4.1. (Conceptual.) Explain why a pendulum swinging on its limit cycle (pumped by a periodic torque) is orbitally stable but *not* Lyapunov stable. Sketch the phase portrait and identify the tangential and transverse directions.

4.2. (Computation.) For the Van der Pol oscillator $\ddot{x} - \mu(1-x^2)\dot{x} + x = 0$ with $\mu = 1$:

(a) Numerically find the limit cycle by simulating from $(x, \dot{x}) = (3, 0)$ until transient effects decay.

(b) Compute the transverse linearization along the limit cycle.

(c) Compute the monodromy matrix and Floquet multipliers numerically.

(d) Verify that one multiplier of the full (2D) system is exactly 1.

4.3. (Transverse linearization.) For the system $\dot{x}_1 = x_2 + u$, $\dot{x}_2 = -x_1^3$, consider the trajectory $\boldsymbol{\gamma}^*(t) = (\cos t, -\sin t)$ with appropriate u^*.

(a) Find $u^*(t)$ such that $\boldsymbol{\gamma}^*$ is a solution.

(b) Compute the full Jacobian $\boldsymbol{J}(t)$ along $\boldsymbol{\gamma}^*$.

(c) Compute the transverse Jacobian $\boldsymbol{A}_\perp(t)$.

(d) Is the orbit stable open-loop? Design stabilizing transverse feedback.

4.4. (Numerical project: Transverse LQR.) For the double pendulum from Exercise 3.4:

(a) Compute a reference swing trajectory using direct collocation (or load from a data file).

(b) Compute the transverse linearization along the trajectory.

(c) Solve the differential Riccati equation with $\boldsymbol{Q}(t)$ that increases 10-fold near the terminal phase (“impact”).

(d) Simulate the closed-loop system with random initial perturbations and plot the transverse deviation $\|\boldsymbol{\xi}(t)\|$ over time.

(e) Compute the amplification ratio profile $\rho(0, s)$ and identify the "natural funnel" phase.

4.5. (Design.) A walking robot with period-1 gait has Floquet multipliers $\mu_1 = 0.9$, $\mu_2 = 1.1$, $\mu_3 = 0.5$.

(a) Is the gait orbitally stable? Which mode is problematic?

(b) Design a transverse feedback controller that places the unstable multiplier μ_2 at 0.6. What gain is required?

(c) After stabilization, what is the worst-case convergence rate per stride?

(d) How many strides does it take to reduce a 10% transverse deviation to 1%?

4.6. (Philosophical.) Compare the transverse LQR approach (this chapter) with the standard trajectory-tracking LQR (linearize about $\boldsymbol{\gamma}^*(t)$, regulate $\boldsymbol{e} = \boldsymbol{x} - \boldsymbol{\gamma}^*(t)$ to zero).

(a) Under what conditions do the two approaches give identical feedback gains? (Hint: consider straight-line trajectories.)

(b) Construct a numerical example where standard LQR uses significantly more control effort than transverse LQR.

(c) Discuss which approach is preferable for the golf swing and why.

The pendulum does not care if you are watching at noon or midnight.
Its orbit knows no clock.

Stability is not arrival. It is the promise
that the motion, once begun, will keep its shape—
even when the world pushes back.

Chapter 5

Underactuation and Passive Dynamics

Key Idea 5.1. How One Input Moves Several Links

Moving the hands can accelerate a club even when a simplified model applies no wrist torque. The club's response depends on its orientation, existing velocity, gravity, and the acceleration of the hands. This is a useful way to study release. It does not establish that a golfer's wrists are freely hinged, that the arms must actively brake, or that the resulting motion is stable or optimal. Those are separate questions that the model and measurements must answer.

5.1 Instantaneous Authority and Motion Over Time

In independent configuration coordinates $q \in \mathbb{R}^n$, write

$$M(q)\ddot{q} + h(q, \dot{q}) = B(q)u, \qquad M = M^T \succ 0. \tag{5.1}$$

Here $h = C(q, \dot{q})\dot{q} + \nabla V(q)$ for an ideal rigid mechanical model; damping, elastic forces, and other loads must be added when retained. At fixed state the attainable acceleration set is

$$\mathcal{A}(q, \dot{q}) = -M^{-1}h + M^{-1}B\mathcal{U}, \tag{5.2}$$

where $\mathcal{U}$ is the admissible input set. With unrestricted inputs its affine dimension is $r = \operatorname{rank} B$. Underactuation means $r < n$. When $m < n$, an $n \times m$ matrix can have full column rank m while lacking full row rank. Multiple muscles do not guarantee independent joint torques: moment arms, force signs, activation, and force limits constrain the attainable set.

Instantaneous authority is different from reachability over time. Coupled dynamics can move an unactuated coordinate. A full-row-rank model also cannot follow every trajectory when contact, torque, power, speed, or state constraints are imposed. A walking foot has no moment at an ideal point contact, whereas distributed foot contact can support a bounded moment. The contact model matters.

5.2 A Coordinate-Consistent Double Pendulum

Let θ_1 and θ_2 be absolute angles of the arm and club measured from downward vertical in one inertial plane. Let τ_s be shoulder torque and τ_w wrist torque exerted on the club. Virtual work gives

$$\delta W = \tau_s \delta\theta_1 + \tau_w(\delta\theta_2 - \delta\theta_1), \qquad Q_\theta = (\tau_s - \tau_w, \tau_w)^T. \tag{5.3}$$

The corresponding actuator power is $\tau_s\dot\theta_1+\tau_w(\dot\theta_2-\dot\theta_1)$. For relative coordinates $q=(\theta_1,\theta_2-\theta_1)^T$, let

$$\theta = Tq, \qquad T=\begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}. \tag{5.4}$$

Then $Q_q=T^TQ_\theta=(\tau_s,\tau_w)^T$ and $M_q=T^TM_\theta T$. For this constant transformation, $h_q=T^Th_\theta$. The shoulder-only map $(1,0)^T$ is valid in both conventions when $\tau_w=0$; the mass and bias terms must still use the chosen coordinates. Wrist power uses relative angular speed.

For a transparent worked model, use massless links of lengths l_1,l_2, point masses m_1 at the wrist and m_2 at the club tip, and positive gravity magnitude g. Set $\Delta=\theta_1-\theta_2$ and $b=m_2l_1l_2$. In absolute angles,

$$M_\theta=\begin{pmatrix}(m_1+m_2)l_1^2 & b\cos\Delta\\ b\cos\Delta & m_2l_2^2\end{pmatrix}, \tag{5.5}$$

$$h_\theta=\begin{pmatrix}b\sin\Delta\,\dot\theta_2^2+(m_1+m_2)gl_1\sin\theta_1\\ -b\sin\Delta\,\dot\theta_1^2+m_2gl_2\sin\theta_2\end{pmatrix}. \tag{5.6}$$

These equations follow from $L=\frac{1}{2}\dot\theta^TM_\theta\dot\theta-V$ with $V=-(m_1+m_2)gl_1\cos\theta_1-m_2gl_2\cos\theta_2$. They omit distributed limb inertia, a moving shoulder, shaft bending, and three-dimensional grip mechanics. Their purpose is an inspectable mechanism.

5.2.1 What Arm Deceleration Does and Does Not Imply

With zero wrist torque, the second row is

$$M_{21}\ddot\theta_1+M_{22}\ddot\theta_2+h_2=0, \qquad \ddot\theta_2=-\frac{M_{21}\ddot\theta_1+h_2}{M_{22}}. \tag{5.7}$$

The positive denominator does not determine the sign of this expression. $M_{21}=b\cos\Delta$ changes sign with geometry, and the velocity and gravity terms remain. With $M_{22}=1$, $h_2=0$, and $\ddot\theta_1=-2$, $M_{21}=0.4$ gives $\ddot\theta_2=0.8$ whereas $M_{21}=-0.4$ gives -0.8. Both matrices with $M_{11}=2$ are positive definite. These are a dimensionally consistent rotational-inertia example, not measured golfer parameters.

Arm deceleration can therefore accompany club acceleration in suitable states. It is not a necessary instruction to actively brake the arms: reaction torques can decelerate an arm while shoulder torque remains positive. Angular momentum of the arm-plus-club system also changes under external shoulder torque and gravity. Calling this an optimal momentum transfer requires an objective, admissible inputs, and an optimization result that these equations alone do not supply.

For the ideal conservative model,

$$\frac{d}{dt}(T_{\text{kinetic}}+V)=\tau_s\dot\theta_1+\tau_w(\dot\theta_2-\dot\theta_1). \tag{5.8}$$

Zero wrist actuator power does not mean zero energy flow into the club. Forces at the moving wrist can do work on the club while doing opposite work on the arm. Whole-system and segment energy budgets answer different questions; include translational boundary power and avoid counting internal power twice.

5.3 The Annihilator Test for Feasible Trajectories

On a constant-rank coordinate patch, let $N_B(q) \in \mathbb{R}^{(n-r)\times n}$ have rows spanning the left nullspace of B. Thus $N_B B = 0$. A twice differentiable configuration path with its stated timing requires the load

$$b_*(t) = M(q_*)\ddot{q}_* + h(q_*, \dot{q}_*). \tag{5.9}$$

The condition

$$N_B(q_*)b_* = 0 \tag{5.10}$$

is equivalent to $b_* \in \operatorname{range} B$: some unrestricted algebraic input produces that load. One solution is $u_* = B^+ b_*$, with possible additions from $\ker B$. The required input must also belong to $\mathcal{U}$. For example, $B = (1,0)^T$ and $b_* = (10,0)^T$ pass the annihilator test but fail an input bound $|u| \leq 1$. Input-rate and actuator-state constraints require additional dynamical checks. A locally smooth constant-rank pseudoinverse does not remove these restrictions.

This is a constraint on position, velocity, and acceleration along a trajectory, not generally a fixed subspace of state space. Changing the timing changes $\dot{q}_*$ and $\ddot{q}_*$, so the same geometric path may pass or fail. Contact reactions must first be eliminated consistently or retained as additional unknowns with their friction and unilateral-contact conditions.

5.4 Partial Feedback Linearization and Zero Dynamics

For an ideal collocated partition with $B = (I, 0)^T$, write

$$M_{aa}\ddot{q}_a + M_{au}\ddot{q}_u + h_a = u, \tag{5.11}$$

$$M_{ua}\ddot{q}_a + M_{uu}\ddot{q}_u + h_u = 0. \tag{5.12}$$

Choosing the acceleration command v_a requires

$$u = (M_{aa} - M_{au}M_{uu}^{-1}M_{ua})v_a + h_a - M_{au}M_{uu}^{-1}h_u. \tag{5.13}$$

The Schur complement is positive definite when $M \succ 0$. Exact cancellation assumes an accurate model and an admissible computed input. Saturation and uncertain loads invalidate the ideal equality $\ddot{q}_a = v_a$.

For output $e_a = q_a - q_a^*(t)$, zero dynamics are the remaining dynamics on $e_a = \dot{e}_a = 0$ under the input that keeps this set invariant. They depend on the output and its reference, and are generally time varying for tracking. They are not identical to the motion obtained by switching all actuators off.

No phase of a generic swing is automatically attracting. Integrate a specified nominal solution and the variational equation

$$\delta\dot{z} = A_z(t)\delta z, \qquad \dot{\Phi}_z = A_z(t)\Phi_z, \quad \Phi_z(t_0) = I. \tag{5.14}$$

Specify the output, comparison time or event, and the metric used to scale angle and velocity errors. Singular values of the appropriately scaled transition map measure finite-time amplification. A time-varying positive metric $W(t)$ satisfying

$$\dot{W} + A_z^T W + W A_z \preceq -2\lambda W \tag{5.15}$$

certifies differential decay on the interval where it holds, with metric bounds needed to translate it into a coordinate norm. Frozen-time eigenvalues and large club speed alone do not supply this certificate. A verified invariant funnel additionally checks its moving boundary, input limits, and disturbances.

5.5 Accessibility Is Not Forward-Time Reachability

For driftless systems $\dot{x} = \sum_i g_i(x)u_i$ with reversible inputs, a bracket-generating distribution on a connected manifold gives the horizontal connectivity described by Chow–Rashevskii. This does not immediately extend to a drifting mechanical system $\dot{x} = f(x) + \sum_i g_i(x)u_i$ with bounded inputs and a fixed downswing duration.

An orbit generated by vector-field flows permits both positive and negative flow times. Forward reachable sets do not reverse an unavoidable drift. For $\dot{x} = 1$, $\dot{y} = u$, $|u| \leq 1$, the fields span the plane, yet from the origin $x(T) = T$ and $|y(T)| \leq T$. No negative x is reachable for $T \geq 0$. Lie-algebra rank can support local accessibility under appropriate regularity and input assumptions; it does not certify global or small-time local controllability. Smooth orbit theorems also require care beyond a pointwise Lie-bracket rank test.

Abnormal extremals have a zero cost multiplier in Pontryagin's principle; singular arcs require separate analysis of the Hamiltonian's input dependence. Neither is diagnosed by visually rapid wrist release. Establish them for a specified optimal-control problem before assigning them a physiological meaning.

5.6 Connecting the Reduced Model to a Real Swing

The useful hypothesis is that proximal motion and earlier work prepare a state from which mechanical coupling contributes to club delivery. Testing it requires measured wrist and hand wrenches, consistent angular coordinates, a moving-base model, and uncertainty estimates. A passive hinge is one comparison model. Compare it with bounded wrist torques and compliant grips on the same task; then assess speed, orientation, dispersion, and effort separately.

Shaft deformation stores energy and changes the contact-point velocity. Muscle activation affects force capacity and impedance. Feedback changes responses to perturbations, while preparation changes the nominal trajectory itself. These mechanisms can coexist. A zero-torque comparison does not identify their unique biological contributions, and a simulated advantage must survive measurement and model-sensitivity checks before becoming a coaching recommendation.

5.7 Exercises

1. Verify the mass and bias terms from the Lagrangian and check the power identity with nonzero wrist torque in both coordinate systems.

2. Reproduce the coupling-sign counterexample. Include gravity and nonzero angular rates and identify when club acceleration changes sign.
3. Construct N_B for a rank-deficient actuator map. Give a load that passes the range test but violates an actuator bound.
4. For $\ddot{q}_u + c q_u = \sin t$, subtract a particular solution. Classify homogeneous perturbations for $c > 0$, $c = 0$, and $c < 0$; examine resonance at $c = 1$. Explain why bounded perturbations do not guarantee bounded forced motion.
5. Specify an experiment that distinguishes passive-hinge prediction error from measurement noise, wrist actuation, and shaft compliance.

Chapter 6

Trajectory Optimization

We do not merely seek a path that obeys the laws of physics.
We seek the path that bends them to our advantage.

6.1 The Moving Target Problem

In Chapter 1, we fundamentally reframed our control objective from "reaching a destination" to "tracking a trajectory." Up to this point, we have assumed that this optimal reference trajectory $\boldsymbol{\gamma}^*(t)$ is given to us by an oracle. This chapter answers the obvious next question: *Where does this reference come from?*

For systems whose purpose is to move—like our running example of the golf swing—the optimal trajectory is not arbitrary. We are not just trying to go from point A to point B while minimizing energy. We are trying to fulfill what we term the **Moving Target Objective**: optimizing the kinematics (velocity, pose, acceleration) of a system precisely at the moment it intersects a target manifold (e.g., the golf ball).

Crucially, *the exact clock time that impact occurs does not matter*. A golfer has the freedom to start their backswing slowly and pause at the top. The optimization is entirely concerned with the *shape* of the motion and its terminal kinematics. Time is a resource, not a constraint.

Definition 6.1. The Moving Target Optimal Control Problem (OCP)

Find the state trajectory $\boldsymbol{x}(t)$, control input $\boldsymbol{u}(t)$, and terminal time t_f that minimize the cost functional:

$$J = \phi(\boldsymbol{x}(t_f)) + \int_0^{t_f} L(\boldsymbol{x}(t), \boldsymbol{u}(t))\,\mathrm{d}t \tag{6.1}$$

subject to the dynamics and constraints:

$$\begin{aligned}
\dot{\boldsymbol{x}}(t) &= \boldsymbol{f}(\boldsymbol{x}(t), \boldsymbol{u}(t)) && \text{(Dynamics)} && (6.2)\\
\boldsymbol{h}(\boldsymbol{x}(t), \boldsymbol{u}(t)) &\leq \boldsymbol{0} && \text{(Path constraints)} && (6.3)\\
\psi(\boldsymbol{x}(t_f)) &= \boldsymbol{0} && \text{(Terminal manifold/Moving goal)} && (6.4)\\
\boldsymbol{x}(0) &= \boldsymbol{x}_0 && \text{(Initial state)} && (6.5)
\end{aligned}$$

In the golf downswing, the terminal cost $\phi(\boldsymbol{x}(t_f))$ might be negative clubhead speed (to maximize speed), the integral cost L might penalize excessive joint torques to ensure repeatability, and the terminal manifold constraint $\psi(\boldsymbol{x}(t_f)) = \mathbf{0}$ forces the clubhead to be geometrically located at the ball with a squared face. Notice that t_f is a *free parameter* in the optimization.

6.2 Transcribing the Continuous Problem

The OCP described above occurs in infinite-dimensional function space. To solve it on a computer, we must transcribe it into a finite-dimensional Nonlinear Programming problem (NLP):

$$\begin{aligned} \min_{\boldsymbol{w}} \quad & f(\boldsymbol{w}) \\ \text{subject to} \quad & \boldsymbol{c}(\boldsymbol{w}) \leq \mathbf{0}, \end{aligned}$$

where $\boldsymbol{w}$ is a large vector of decision variables. How we discretize the integrals and differential equations defines the transcription method. We will focus on two modern approaches: **Direct Shooting** and **Direct Collocation**.

6.2.1 Direct Multi-Shooting

In direct multi-shooting, we break the time interval $[0, t_f]$ into N segments. The decision variables are the control parameters $\boldsymbol{u}_k$ on each interval and the state values $\boldsymbol{x}_k$ at the beginning of each interval.

The dynamic constraint $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})$ is enforced by numerical integration (e.g., Runge-Kutta). We require that the integrated state at the end of interval k exactly matches the assumed starting state of interval $k+1$:

$$\boldsymbol{x}_{k+1} - \boldsymbol{\Phi}(\boldsymbol{x}_k, \boldsymbol{u}_k, \Delta t) = \mathbf{0} \qquad \text{(Defect constraint)} \tag{6.6}$$

where $\boldsymbol{\Phi}$ represents the numerical integrator step. Multi-shooting is excellent for highly nonlinear dynamics because it keeps the integration intervals short, preventing the massive nonlinear sensitivities that plague single-shooting methods.

6.2.2 Direct Collocation

Direct Collocation takes this a step further. We represent both the state trajectory and the control signal as piecewise polynomials (e.g., cubic splines for state, linear splines for control). The decision variables are the coefficients of these polynomials, typically located at discrete **collocation points**.

Instead of using a numerical integrator, we enforce the differential equation algebraicaly at the collocation points:

$$\dot{\tilde{\boldsymbol{x}}}(t_c) - \boldsymbol{f}(\tilde{\boldsymbol{x}}(t_c), \tilde{\boldsymbol{u}}(t_c)) = \mathbf{0} \qquad \text{(Collocation constraint)} \tag{6.7}$$

where $\tilde{\boldsymbol{x}}$ and $\tilde{\boldsymbol{u}}$ are the polynomial approximations.

Key Idea 6.1. Fidelity vs. Flexibility

Direct collocation results in a massive, highly sparse NLP. Because the solver has access to the state at every grid point simultaneously, it can easily manipulate the physical trajectory to satisfy difficult path constraints without waiting to "simulate" the result. Collocation is widely considered the state-of-the-art for generating complex dynamic motions like walking, jumping, and swinging.

6.3 Optimizing Underactuated Geometries

Recall from Chapter 5 that the underactuated dynamics enforce a strict algebraic constraint on the system:

$$\boldsymbol{B}^{\perp}(\boldsymbol{x})\Big[\boldsymbol{M}(\boldsymbol{x})\ddot{\boldsymbol{x}} + \boldsymbol{C}(\boldsymbol{x},\dot{\boldsymbol{x}})\dot{\boldsymbol{x}} + \boldsymbol{g}(\boldsymbol{x})\Big] = \boldsymbol{0}. \tag{6.8}$$

When optimizing for a golf swing, direct collocation handles this underactuation organically. We do not need to explicitly partition the variables into actuated and unactuated sets for the solver. The collocation constraint $\dot{\boldsymbol{x}} - \boldsymbol{f} = \boldsymbol{0}$ already implicitly contains the passive dynamic constraints.

Example 6.1. The Golf Swing OCP

Consider our planar double pendulum model from Chapter 5. The solver is given the task to maximize $\dot{\theta}_2(t_f)$ (club speed).
Initial Guess: If we seed the NLP solver with a naive guess (e.g., constant shoulder torque), the club speed will be substantially suboptimal.
The Optimized Result: The solver discovers the optimal physical orchestration entirely on its own. It will apply a massive positive u_1 to accelerate the arms, and then immediately reverse u_1 to negative maximum just prior to t_f. The optimizer mathematically "discovers" the necessary inertial coupling (the parametric excitation of $M_{21}\ddot{\theta}_1$) to whip the unactuated club link through the terminal manifold constraint! The passive dynamics are beautifully exploited to satisfy the objective.

6.4 Repeatability: The Stochastic OCP

The previous sections frame a deterministic problem: "Find the swing that maximizes terminal speed."

But as we argued in Chapter 1, biological systems and athletes do not optimize for peak deterministic performance. If you instruct an NLP solver to maximize golf club speed without regard for motor noise, the solver will ride the very edge of the constraint boundaries. It will produce a swing so delicately timed that a 5-millisecond execution error would result in a complete whiff.

Principle 6.1. The Repeatability Addendum

The theoretically optimal trajectory is useless if its local topology amplifies small execution errors into catastrophic terminal misses. The true objective must account for *variance.*

We modify our deterministic cost $J = \phi(\boldsymbol{x}(t_f)) + \int L$ into a stochastic expectation:

$$\min_{\boldsymbol{\gamma}^*} \quad \mathbb{E}\left[\phi(\boldsymbol{x}(t_f, \boldsymbol{w}))\right] + \lambda \cdot \mathrm{Var}\left[\psi(\boldsymbol{x}(t_f, \boldsymbol{w}))\right] \tag{6.9}$$

where $\boldsymbol{w} \sim \mathcal{N}(0, \Sigma_w)$ represents signal-dependent motor noise (a known property of human muscle actuation, where noise scales proportionally to the applied torque).

To solve this in a collocation framework without massive Monte Carlo simulations, we use **Covariance Steering** or **Linear Covariance Analysis**. We augment the state vector $\boldsymbol{x}(t)$ with its local covariance matrix $\boldsymbol{\Sigma}(t)$. The dynamics of $\boldsymbol{\Sigma}(t)$ are propagated alongside the nominal trajectory using the linearized Jacobian matrices along the path.

The optimizer is now forced to "widen the funnel" as it approaches the moving goal. It sacrifices a small amount of peak clubhead speed to select a trajectory geometry where the passive dynamics are heavily stabilizing in the transverse direction. The result is a trajectory that a human can actually *repeat.*

6.5 Summary

Concept	**Significance**
Moving Target OCP	Finding a trajectory that optimizes terminal kinematics when intersecting a spatial manifold, without prescribing the clock time.
Multi-Shooting	Discretizing the time domain into segments, using numerical integrators to enforce dynamics constraints between nodes.
Direct Collocation	Representing the entire trajectory as splines and enforcing dynamics algebraically at discrete points arrayed along the trajectory.
Passive Exploitation	Solvers naturally optimize underactuated topologies by exploiting coupling matrices (M_{21}, C_2).
Stochastic Robustness	Augmenting the OCP with expected variance to ensure the resulting trajectory can be consistently executed by noisy nonlinear plants.

Determining the shape of the reference trajectory is the cornerstone of Control Is Motion. With our passive geometry understood (Chap 5) and our optimal reference trajectory computed (Chap 6), we must now surround that reference trajectory with a guaranteed, robust control tube to reject disturbances online. We turn to Funnel Synthesis in Chapter 7.

6.6 Exercises

6.1. **(Free Terminal Time Formulation.)** Explain how one sets up an OCP where final time t_f is a decision variable in a fixed-grid Direct Collocation solver. (Hint: consider scaling time $t = \tau \cdot t_f$, where $\tau \in [0, 1]$). Show how the dynamic constraint $\dot{\boldsymbol{x}} = \boldsymbol{f}$ changes under this substitution.

6.2. **(Kinematic Goals vs Setpoints.)** You are optimizing a baseball bat swing. Your state is $(\theta, \dot{\theta})$. Assume the ball arrives at the plate between $t = 0.5$ and $t = 0.6$ seconds. Formulate the path constraint and the terminal manifold constraint ψ assuming you want maximum $\dot{\theta}$ when $\theta = \pi/2$, but the exact time of impact within that window does not matter.

6.3. **(Signal-Dependent Noise Penalty.)** Human muscle torque output τ has variance $\sigma^2 = k|\tau|^2$. Write a modified integral cost function $L(\boldsymbol{x}, \tau)$ that penalizes the introduction of variance into the system. Interpret what this does to the optimal trajectory shape.

The perfect motion is rarely the fastest.
It is the one that survives reality.

Chapter 7

Funnel Synthesis

You cannot control the exact trajectory a stone will take down a mountain.
But you can build a trough to ensure it reaches the bottom.

7.1 Beyond LQR: The Need for Guaranteed Invariance

In Chapter 4, we showed that applying time-varying Linear Quadratic Regulation (LQR) to the transverse dynamics creates a contracting envelope of stability. The Riccati equation naturally shapes this envelope into a "funnel" that narrows as the terminal constraint approaches.

However, LQR guarantees are inherently *local*. They rely on the linear approximation of the transverse dynamics $\dot{\boldsymbol{\xi}} = \boldsymbol{A}_\perp(t)\boldsymbol{\xi} + \boldsymbol{B}_\perp(t)\delta\boldsymbol{u}$. For actual physical systems traversing highly nonlinear state spaces—like a golf club accelerating to 50 m/s while subject to massive centrifugal and Coriolis forces—this linearization quickly breaks down. If a disturbance pushes the state too far from the reference trajectory $\boldsymbol{\gamma}^*(t)$, the nonlinearities will dominate, the LQR restoring forces will compute the wrong corrective action, and the clubhead will completely miss the ball.

We need a mathematically rigorous way to define the *exact boundary* of our funnel. We must prove that for the fully nonlinear equations of motion, any state starting within the mouth of the funnel at time $t = 0$ will remain inside the funnel for all $t \in [0, t_f]$.

Definition 7.1. Dynamic Funnel

Given a reference trajectory $\boldsymbol{\gamma}^*(t)$, a feedback controller $\boldsymbol{u} = \boldsymbol{k}(\boldsymbol{x}, t)$, and a scalar boundary function $V(\boldsymbol{x}, t)$, the set:

$$\mathcal{F}(t) = \{\boldsymbol{x} \in \mathcal{X} \mid V(\boldsymbol{x}, t) \leq 1\} \tag{7.1}$$

constitutes a valid **dynamic funnel** if the set is positively invariant under the closed-loop dynamics. That is, if $\boldsymbol{x}(0) \in \mathcal{F}(0)$, then $\boldsymbol{x}(t) \in \mathcal{F}(t)$ for all $t > 0$.

7.2 Time-Varying Lyapunov Functions for Trajectories

To verify invariance, we use the machinery of Lyapunov theory, adapted for trajectories rather than equilibria. We require our boundary function $V(\boldsymbol{x}, t)$ to act as a time-varying Lyapunov function.

Theorem 7.1 (Funnel Invariance)**.** *The set $\mathcal{F}(t) = \{\boldsymbol{x} \mid V(\boldsymbol{x}, t) \leq 1\}$ is invariant if, on the boundary where $V(\boldsymbol{x}, t) = 1$, the function is strictly decreasing along the system trajectories:*

$$\dot{V}(\boldsymbol{x}, t) = \frac{\partial V}{\partial t} + \nabla_{\boldsymbol{x}} V(\boldsymbol{x}, t) \cdot \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{k}(\boldsymbol{x}, t)) < 0 \quad \textit{for all } \boldsymbol{x} \textit{ s.t. } V(\boldsymbol{x}, t) = 1. \tag{7.2}$$

If the derivative of the boundary function points inward everywhere on the boundary, trajectories may circulate inside the funnel, but they can never escape it. For a moving target problem, we want the spatial volume of $\mathcal{F}(t)$ to shrink as $t \to t_f$, terminating in a tight bounded region at impact.

Key Idea 7.1. Funnels Define the Usable State Space

A computed funnel is not just an analysis tool; it is the ultimate judge of repeatability. It explicitly defines the exact set of initial conditions (e.g., the kinematic state at the top of the backswing) that are mathematically guaranteed to successfully strike the ball despite the presence of extreme nonlinearities.

7.3 Sums-of-Squares (SOS) Programming

Proving that $\dot{V} < 0$ on the boundary $V = 1$ for a general nonlinear function $\boldsymbol{f}$ is famously difficult. We have an infinite number of states on the boundary to check! We circumvent this by restricting our dynamics $\boldsymbol{f}$ algorithms to polynomials (via Taylor expansion or substitution) and leveraging **Sums-of-Squares (SOS) Programming**.

A polynomial $p(x)$ is a Sum of Squares if it can be written as $p(x) = \sum_i q_i(x)^2$ for some polynomials $q_i(x)$. If a polynomial is SOS, it is globally non-negative. Testing if a polynomial is SOS turns out to be equivalent to solving a Semidefinite Program (SDP)—a convex optimization problem that can be solved very efficiently.

We can reframe Theorem 7.1 using the S-procedure. To guarantee that $\dot{V}(\boldsymbol{x}, t) < 0$ whenever $V(\boldsymbol{x}, t) = 1$, we require:

$$-\dot{V}(\boldsymbol{x}, t) - L(\boldsymbol{x}, t)(1 - V(\boldsymbol{x}, t)) \quad \text{is SOS} \tag{7.3}$$

where $L(\boldsymbol{x}, t)$ is a strictly positive polynomial multiplier. If $V(\boldsymbol{x}, t) = 1$, the second term vanishes, and the SOS condition forces $-\dot{V} > 0$, achieving our invariance proof!

7.4 Computing the Funnel for the Golf Swing

Let us return to the double pendulum of the golf swing. Our reference trajectory $\boldsymbol{\gamma}^*(t)$ from Chapter 6 gives us the nominal kinematics. Our transverse LQR from Chapter 4 gives us a feedback matrix $\boldsymbol{K}(t)$ designed to stabilize deviations. We wish to compute the largest possible funnel around this sequence that the closed-loop system is mathematically guaranteed to stay inside.

We define our candidate Lyapunov function as a time-varying quadratic form:

$$V(\boldsymbol{x}, t) = (\boldsymbol{x} - \boldsymbol{\gamma}^*(t))^\top \boldsymbol{P}(t)(\boldsymbol{x} - \boldsymbol{\gamma}^*(t)) \tag{7.4}$$

where $\boldsymbol{P}(t)$ is a positive definite matrix that changes continuously along the trajectory. The funnel boundary is the time-varying ellipsoid where $V = 1$. By altering $\boldsymbol{P}(t)$, we alter the shape and orientation of the funnel.

Example 7.1. Volume Maximization via SDP

We transcribe the continuous time t into discrete nodes t_k, just as we did in direct collocation. At each node, we set up an SDP that seeks to *maximize* the volume of the ellipsoid defined by $\boldsymbol{P}(t_k)$, subject to the SOS invariance constraints connecting it to adjacent nodes.

The optimizer actively computes the exact structural limits of the golf swing's passive dynamics. It discovers that early in the downswing, the funnel can be incredibly "wide" (forgiving), but as centrifugal forces build, the SOS constraints force $\boldsymbol{P}(t_k)$ to become highly elongated along the unactuated dimensions in order to maintain the invariance guarantee. The resulting geometry is a true numerical portrait of athletic consistency.

In Chapter 8, we will explore how to make these funnels robust to temporal variation by migrating out of the time domain entirely and into the phase domain.

Chapter 8

Phase-Variable Control

Time is a river that carries us forward. Phase is the boat we build to navigate it.

8.1 Spatial Paths vs. Temporal Execution

One common limitation of direct time-tracking architectures, when applied to biomechanics or advanced robotics, is their rigid adherence to the clock. If you construct a time-varying trajectory $\boldsymbol{\gamma}^*(t)$ for a robotic arm and command a tracking controller to follow it, the controller evaluates the error as $\boldsymbol{e}(t) = \boldsymbol{x}(t) - \boldsymbol{\gamma}^*(t)$.

If the robot hits an unmodeled friction patch and slows down, the reference trajectory $\boldsymbol{\gamma}^*(t)$ continues advancing. The controller can calculate a large position error not because the robot has left the intended path, but because it is behind schedule. Depending on gains, actuator limits, and saturation handling, this can produce excessive catch-up commands.

Key Idea 8.1. The Autonomy of Time

A system executing a motion can often be controlled more safely when progress is measured by state-dependent phase rather than by clock time alone. If the system is impeded, the reference can slow down. The mathematical task is to decouple the spatial geometry of the motion from its temporal execution by replacing chronological time t with a locally monotone **phase variable** s.

Instead of parameterizing our optimal trajectory as $\boldsymbol{\gamma}^*(t)$, we parameterize it as a function of the phase: $\boldsymbol{\gamma}^*(s)$. The phase $s \in [0, 1]$ represents the percentage of motion completed. For a walking robot, s could be the forward position of the hip. For a golf swing, s could be the angular position of the shoulder relative to the ball. The reference trajectory is now a spatial curve with a separately modeled timing law.

This construction is publication-safe only after the phase variable's domain, monotonicity condition, and reset behavior are specified.

Definition 8.1. Phase Variable

A phase variable $s(\boldsymbol{x})$ is a continuously differentiable scalar function of the state $\boldsymbol{x}$ such that its time derivative is strictly positive along all feasible trajectories inside the nominal funnel:

$$\dot{s}(\boldsymbol{x}) > 0 \quad \text{for all } \boldsymbol{x} \in \mathcal{F}. \tag{8.1}$$

8.2 Virtual Constraints

If the goal is no longer to match a time sequence but rather to conform to a spatial path defined by $s(\boldsymbol{x})$, we can express the control objective as a set of holonomic constraints on the state. However, because these constraints are not mechanical linkages but rather enforced by software, they are termed **Virtual Constraints**.

Let our control objective be to drive the state $\boldsymbol{x}$ to the invariant manifold $\mathcal{Z}$ defined by:

$$\mathcal{Z} = \{\boldsymbol{x} \mid \boldsymbol{x} - \boldsymbol{\gamma}^*(s(\boldsymbol{x})) = \mathbf{0}\}. \tag{8.2}$$

By using partial feedback linearization (Chapter 5), we can design a synthetic input $\boldsymbol{v}_a$ that drives the actuated degrees of freedom $\boldsymbol{q}_a$ onto their nominal surfaces proportional to the phase:

$$y = \boldsymbol{q}_a - \boldsymbol{\gamma}_a^*(s(\boldsymbol{x})) \to 0. \tag{8.3}$$

As $y \to 0$, the system converges toward the virtual constraints. Crucially, the rate at which it progresses along the path is entirely governed by $\dot{s}$, which is itself determined by the current momentum and state of the system, not a stopwatch.

8.3 Hybrid Zero Dynamics

When the virtual constraints are perfectly satisfied ($y = 0$), the entire complexity of the n-dimensional state space collapses down into a highly reduced dynamical system that evolves strictly along the manifold $\mathcal{Z}$. This is the **Zero Dynamics** of the system.

For systems that undergo discrete impacts (a foot hitting the ground, or a golf club striking a ball), the system evolves via continuous differential equations until an impact surface is triggered, followed by an instantaneous jump map (a reset) dictated by conservation of momentum.

Principle 8.1. Hybrid Zero Dynamics (HZD)

A control architecture achieves Hybrid Zero Dynamics if the invariant manifold $\mathcal{Z}$ (the zero dynamics surface) is invariant not just under the continuous flow of the differential equations, but also invariant under the discrete impact map. That is, an impact starting on the manifold $\mathcal{Z}$ must immediately map to a new state that is also on $\mathcal{Z}$.

By enforcing HZD, we can design rhythmic walking gaits that are astonishingly robust. If the robot trips, its phase slows down, but the virtual constraints remain active, keeping the limbs on the correct spatial path until the next footfall resets the sequence. The temporal sequence arises naturally from the interplay of gravity and momentum.

8.4 Application: The Golf Downswing

While the golf swing is not a periodic walking gait, a phase-variable approach can still be useful if the selected marker remains monotone through the interval being modeled. A candidate phase might be shoulder angle, hand path progress, or another measured progress coordinate; none is automatically valid outside the stated model. The target event is impact, not a universal manifold $s = 1.0$ independent of the coordinate choice.

By structuring a model through virtual constraints, the analysis can separate spatial deviations from timing delays. A transverse feedback law (Chapter 4) can correct selected spatial errors, while the phase variable schedules arm actuation (u_1) relative to measured progress rather than clock time. This can improve timing robustness, but it does not guarantee perfect passive impact dynamics; impact quality still depends on state estimation, actuator limits, contact modeling, and the validity of the chosen phase coordinate.

Chapter 9

Stochastic Trajectories & Motor Variability

Perfection is inherently fragile. Biological optimization seeks the strongest balance between intent and reality.

9.1 Signal-Dependent Noise

When a roboticist specifies a torque of 50 Nm, the motor delivers 50 Nm with a tiny, constant variance. Biological actuators do not work this way. Human muscles generate force by recruiting discrete motor units. As higher forces are required, larger, more fast-twitch oriented motor units are recruited. This physiological reality manifests as a profound control challenge: **motor noise is proportional to the command signal**.

Definition 9.1. Signal-Dependent Noise (SDN)

A control input $\boldsymbol{u}(t)$ applied by a biological system is actually realized as a stochastic variable $\tilde{\boldsymbol{u}}(t)$:

$$\tilde{\boldsymbol{u}}(t) = \boldsymbol{u}(t) + \boldsymbol{W}\boldsymbol{u}(t) \cdot \boldsymbol{\epsilon}(t) \tag{9.1}$$

where $\boldsymbol{W}$ is a scaling matrix representing the coefficient of variation, and $\boldsymbol{\epsilon}(t) \sim \mathcal{N}(0, I)$ is white noise. That is, the variance of the applied force grows squarely with the magnitude of the ordered force: $\mathrm{Var}(\tilde{u}) = cu^2$.

This simple reality crumbles deterministic optimal control. If you compute an optimal trajectory that requires maximum muscular exertion, the trajectory will mathematically possess minimum variance only if noise is constant. Under SDN, maximum exertion guarantees maximum chaotic noise.

9.2 Minimum Variance Theory of Human Movement

Why do humans move in smooth, bell-shaped velocity profiles rather than jerky "bang-bang" optimal control solutions when reaching for coffee?

Classical control attempts to explain this via "minimum jerk" theories, positing that the nervous system explicitly encodes an optimization functional to minimize $\int \dddot{x}^2 \, dt$. The trajectory-first perspective offers a much deeper, physically grounded explanation championed by Harris and Wolpert: the **Minimum Variance Theory**.

Key Idea 9.1. Optimization for Repeatability

The human nervous system seeks to move in a way that minimizes the final endpoint variance evaluated across multiple trials.

Under SDN, the only way to minimize endpoint variance (ensure you actually grasp the coffee cup smoothly) is to minimize sudden spikes in required actuator forces, because large forces inject huge amounts of uncertainty into the dynamics. Smooth trajectories inherently possess lower peak torque commands than rapid accelerations and decelerations. Smoothness is not the explicitly programmed objective; it is an emergent property of attempting to be consistently accurate under the unavoidable reality of signal-dependent noise.

9.3 The Speed-Accuracy Tradeoff in the Golf Swing

Fitts's Law states that the faster you try to move to a target, the less accurate you become. In golf, we see the supreme manifestation of this. The deterministic optimal control solution (Chapter 6) tells the solver to maximize input torque to achieve theoretically maximal clubhead speed.

However, if we introduce the Stochastic OCP (via Covariance Steering) from Chapter 6 and apply the realistic premise of SDN, the optimizer faces a profound tradeoff.

Applying maximum u_1 torque during the downswing injects enormous variance into the kinematic state. The unactuated club (θ_2) begins whipping through its zero dynamics, but its exact phase progression is now highly uncertain due to the noise flooding the arms.

To ensure the clubface squares up at the moving goal, the stochastic optimizer will *deliberately back off* the peak requested torque. It trades a deterministic 5 mph gain in clubhead speed for a massive reduction in angular variance at the target manifold. The optimal biological swing is mathematically proven to be a sub-maximal physical exertion.

Chapter 10

Learning to Move

The first swing establishes the manifold. The ten-thousandth swing perfects the funnel.

10.1 Iterative Learning Control (ILC)

Trajectory optimization (Chapter 6) computes a reference motion strictly from a mathematical model. However, dynamic models of humans—and even robots—are invariably flawed. Frictional forces are unmodeled, inertias are estimated, and actuators possess hidden dynamics. If you execute a computed reference $\boldsymbol{\gamma}^*(t)$ open-loop, the physical system will likely fail to strike the moving target due to these discrepancies.

Iterative Learning Control (ILC) resolves this by leveraging the repetitive nature of tasks like a golf swing. ILC exploits the premise that while models are inaccurate, *the unmodeled dynamics are highly deterministic across repeated trials.*

If a robot under-swings by 5 cm on the first attempt, the controller records this terminal error and updates the feedforward command $\boldsymbol{u}_{\text{ff}}$ for the *second* attempt. Over repeated trials, the system learns the inverse dynamics of the true physical plant without ever explicitly identifying the model parameters.

Definition 10.1. ILC Update Law

Let j denote the trial index. The feedforward command for trial $j+1$ is updated based on the command and the *transverse error* $\boldsymbol{\xi}$ from trial j:

$$\boldsymbol{u}_j(t) = \boldsymbol{u}_{\text{ff},j}(t) + \boldsymbol{K}_\perp(t)\boldsymbol{\xi}_j(t) \tag{10.1}$$

$$\boldsymbol{u}_{\text{ff},j+1}(t) = \boldsymbol{u}_{\text{ff},j}(t) + \boldsymbol{L}(t)\boldsymbol{\xi}_j(t) \tag{10.2}$$

where $\boldsymbol{L}(t)$ is a learning filter. This updates the nominal trajectory to iteratively cancel out deterministic disturbances.

10.2 Policy Search and Reinforcement Learning

While ILC updates the feedforward commands to zero-out deterministic tracking errors, **Policy Search** algorithms optimize the *shape of the funnel* itself.

Instead of computing the transverse LQR gains $\boldsymbol{K}_\perp(t)$ from an analytical Riccati equation (Chapter 4), we parameterize the controller as a policy $\pi_{\boldsymbol{\theta}}(\boldsymbol{x})$ and search the parameter space $\boldsymbol{\theta}$ to minimize the variance at the moving goal under realistic noise.

Reinforcement Learning (RL), specifically policy gradient methods, allows the agent to sample thousands of golf swings in simulation, continuously updating its policy parameters to maximize the expected reward (clubhead speed and squareness at impact). A heavily trained RL agent organically converges on the very principles we mathematically derived in earlier chapters:

1. It abandons rigid time-tracking and autonomously discovers phase-variable mapping.
2. It exploits the underactuated double-pendulum whipping motion to gain speed.
3. It artificially widens its "funnel" during the transition phase, realizing that overly tight feedback control here merely amplifies motor noise and spoils the impact constraints.

10.3 Trajectory Libraries

A singular optimal trajectory $\boldsymbol{\gamma}^*$ and its corresponding funnel $\mathcal{F}$ are highly specific to a single objective. What if the golfer wishes to fade the ball, hit a draw, or execute a punch shot out of the rough?

Instead of re-running a massive NLP optimization for every scenario, biological systems and advanced controllers maintain **Trajectory Libraries**. A library consists of several distinct, pre-computed optimal trajectories and their corresponding local funnels.

When faced with a new task (e.g., striking the ball on an uneven lie), the system searches its library for the closest matching funnel. By interpolating between established funnels using techniques like LQR-Trees or simple spatial blending, the system can instantly generate a stable, feasible motion without any online optimization.

Chapter 11

Case Study: The Complete Golf Swing

The physics are invariant. What separates the professional is their mastery over the geometry of the resultant funnel.

11.1 Unifying the "Control Is Motion" Framework

Throughout this text, we have systematically deconstructed the classical control paradigm and rebuilt it to handle tasks defined uniquely by motion. We conclude by applying our holistic framework to a 15-DOF musculoskeletal model of the human golf swing synthesizing the mathematical structure underlying skilled athletic motion.

The objective is clear: maximize negative clubhead velocity at the precise spatial intersection with the golf ball, while maintaining face angle orthogonality. The exact time interval t_f in which this occurs is irrelevant; only the geometric state at impact matters. Furthermore, the commanded motion must not exceed the structural limits of biological torque, and the trajectory must be highly robust to physiological signal-dependent noise (SDN).

11.1.1 Phase 1: Defining the Geometry and Optimization

The system is heavily underactuated. The wrists are treated as passive pin joints, delegating the final propulsive phase entirely to inertial coupling.

We formulate a **Stochastic Moving Target Optimal Control Problem** (Chapter 6). We deploy Direct Collocation to optimize the trajectory $\boldsymbol{\gamma}^*(s)$ parameterized over a spatial phase variable s representing the shoulder angle (Chapter 8). The integral cost function strictly penalizes signal variance (Chapter 9) by incorporating local covariance propagation.

The collocation solver successfully navigates the $\boldsymbol{B}^{\perp}$ annihilator constraint (Chapter 5) and produces an optimal, smooth backswing. It discovers that a massive torque inversion (braking the torso) right before impact causes the zero dynamics of the passive wrist to wildly accelerate, perfectly squaring the face precisely as the clubhead spatial coordinate intersects the ball's location.

11.1.2 Phase 2: Funnel Synthesis and Robustness

With $\boldsymbol{\gamma}^*(s)$ computed, we lock down the trajectory using **Transverse Linearization** (Chapter 4). We ignore errors along the tangent of the path (since chronological timing is irrelevant) and construct a time-varying LQR feedback matrix to regulate only the transverse deviations.

Finally, we compute the verified bounds of this stability region using **Sums-of-Squares Programming** (Chapter 7). The SDP calculates a rigorous boundary function $V(\boldsymbol{x}, t) = 1$ that proves the system is confined to an invariant funnel.

Key Idea 11.1. The Architecture of the Perfect Swing

The final computed funnel reveals the mathematical secret of the professional golfer. During the backswing, the funnel is immensely wide; the motion is forgiving, and vast transverse deviations are easily tolerated without violating the invariant boundary. As the swing approaches the transition and the explosive downswing, the funnel aggressively tightens. The stochastic optimization has selected a trajectory whose passive unactuated dynamics are so profoundly stable in the final 0.15 seconds that the funnel diameter at impact shrinks to virtually zero. Any small deviation injected by muscular noise is passively swallowed by the immense centrifugal geometry of the zero dynamics. The golfer does not aggressively steer the club to the ball; they simply set the initial conditions, and the physics irresistibly pull the clubface through the target manifold.

Through this book, we hope we have proven that the thermostat has its place, but when the objective is a sweeping, dynamic motion cutting through the physical world, the trajectory-centric perspective of "Control is Motion" provides a natural and expressive mathematical language.